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ONERA



THE FRENCH AEROSPACE LAB

www.onera.fr

Bayesian Optimization: mixed integer and multiobjective applications

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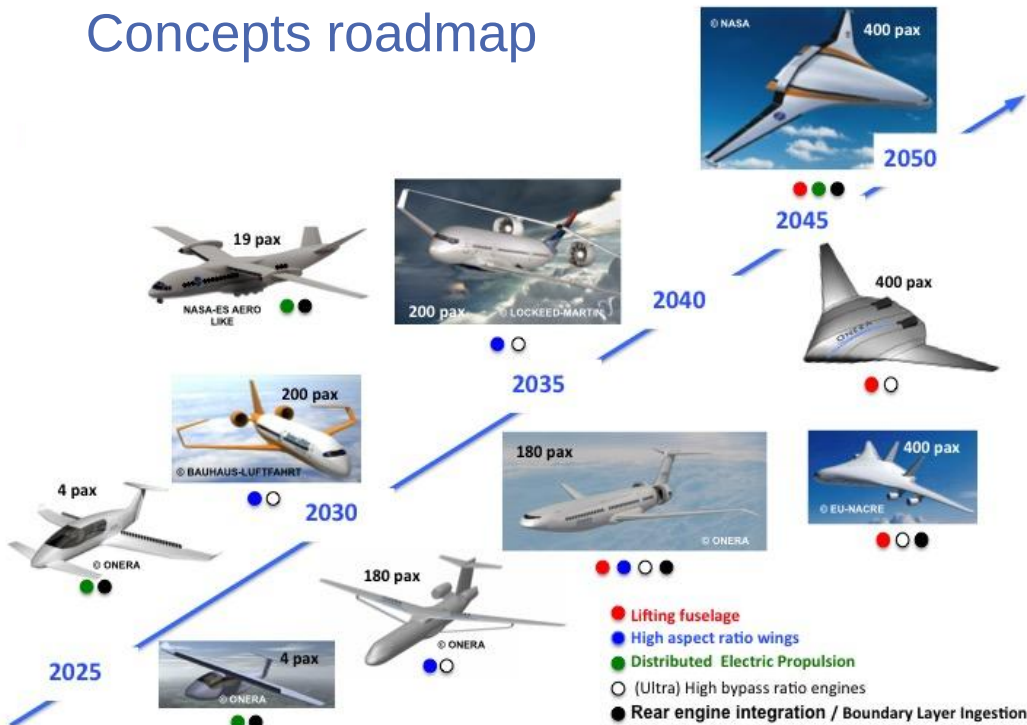


Youssef Diouane
Joseph Morlier



Tools to design the new aircraft configurations

Concepts roadmap



MDO Framework

Surrogate models

Disciplinary Tools

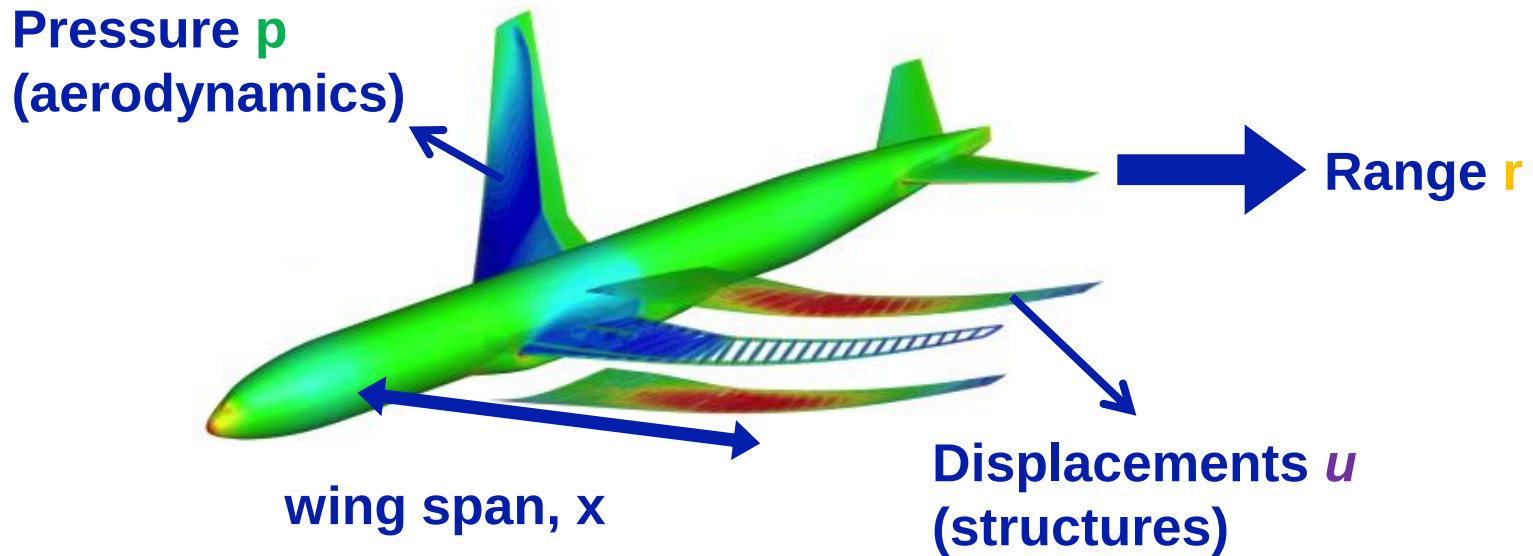
Optimization Tools

Uncertainty Quantification Tools

- ➔ Take into account more information earlier in the design process
- ➔ Keep a design space as large as possible

State of the art: Multidisciplinary Design Analysis and Optimization

Goal: Aircraft/drone design optimization

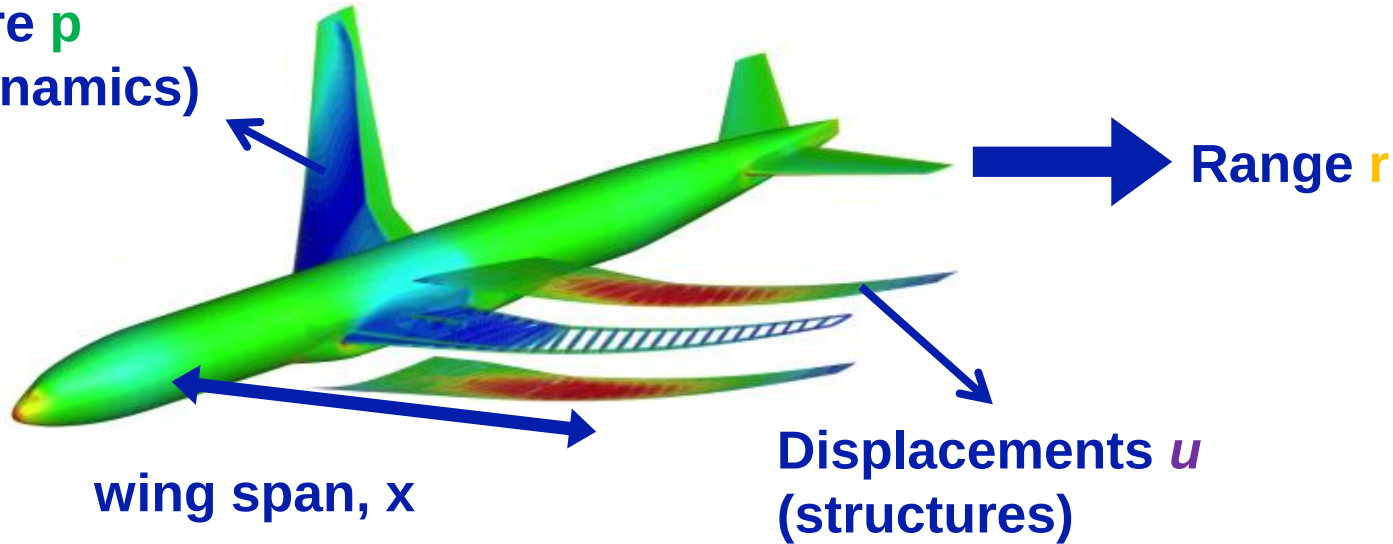


Joaquim R. R. A. Martins and Andrew Ning. Engineering Design Optimization. Cambridge University Press, 2020.

State of the art: Multidisciplinary Design Analysis and Optimization

Goal: Aircraft/drone design optimization

Pressure p
(aerodynamics)



min objective function $f(x, y(x))$
with respect to design variables x
subject to constraints

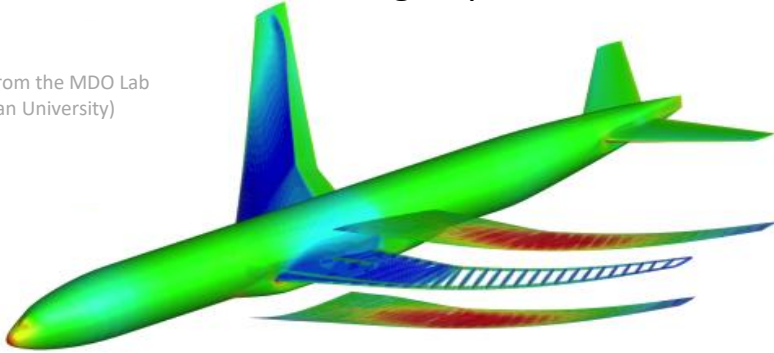
→ $y(x)$ are the state variables (displacements and pressure), solution of a non linear system (MDA) $R(x, y(x))=0$

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State of the art: Multidisciplinary Design Analysis and Optimization

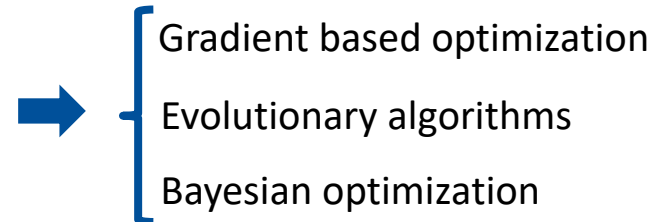
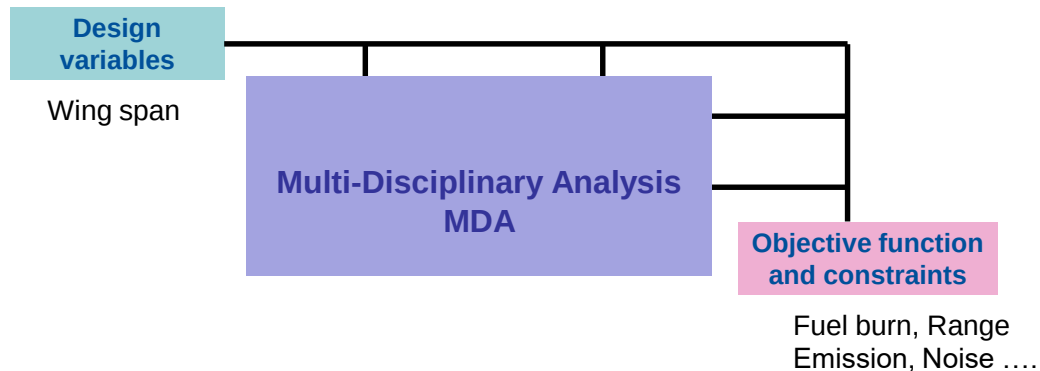
Goal: Aircraft/drone design optimization

Image from the MDO Lab
(Michigan University)



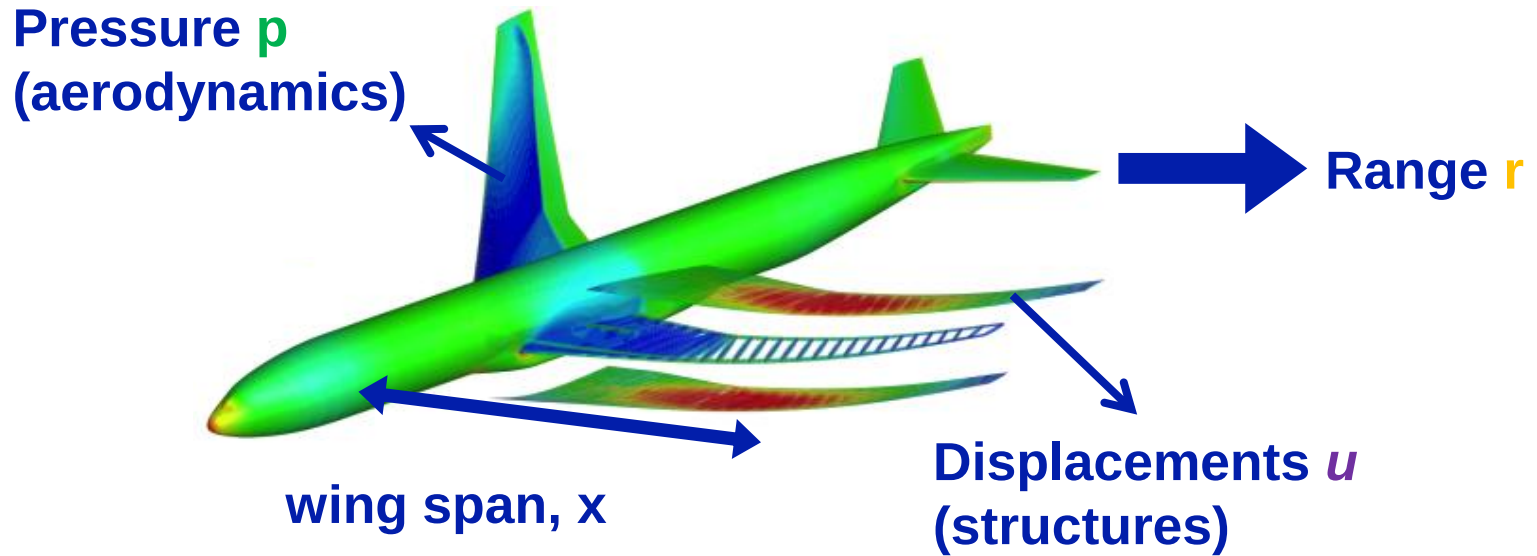
MDO needs:

- A non-linear solver (Gauss-Seidel, Newton,...) to solve MDAs
- An optimizer (gradient based, gradient free, ...) to solve the optimization problem



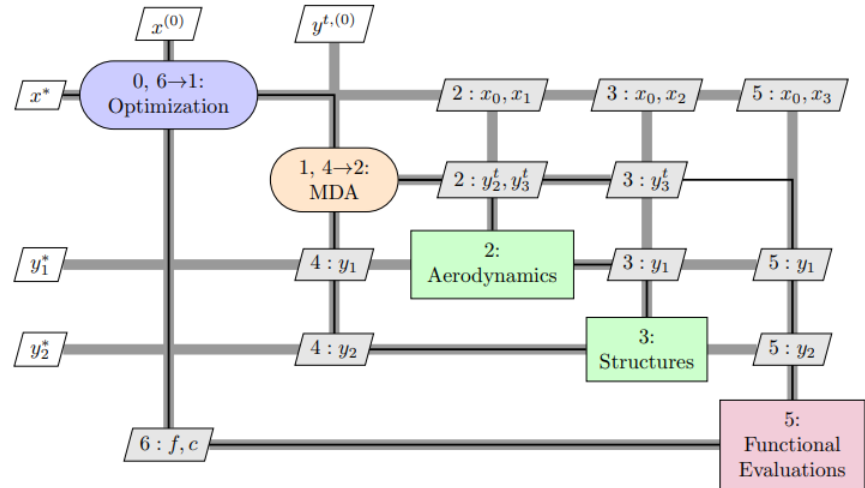
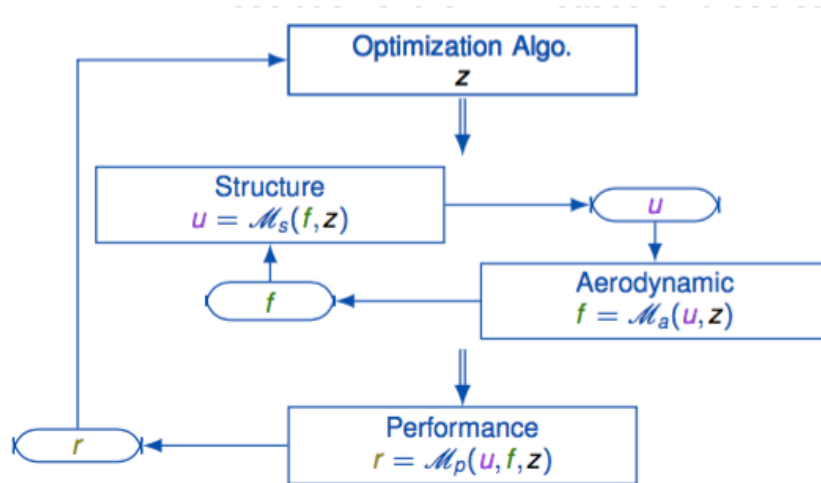
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Different strategies



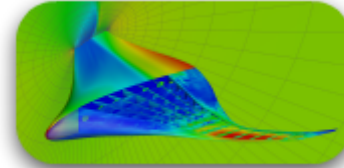
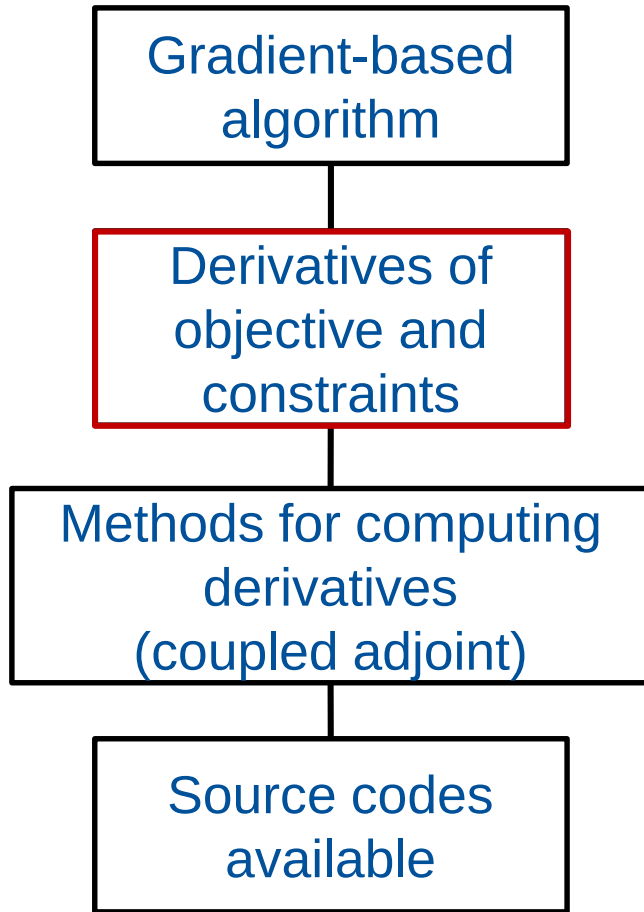
- Lambe, A. B., & Martins, J. R. (2012). Extensions to the design structure matrix for the description of multidisciplinary design, analysis, and optimization processes. *Structural and Multidisciplinary Optimization*, 46(2), 273-284.
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Different strategies



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Different strategies

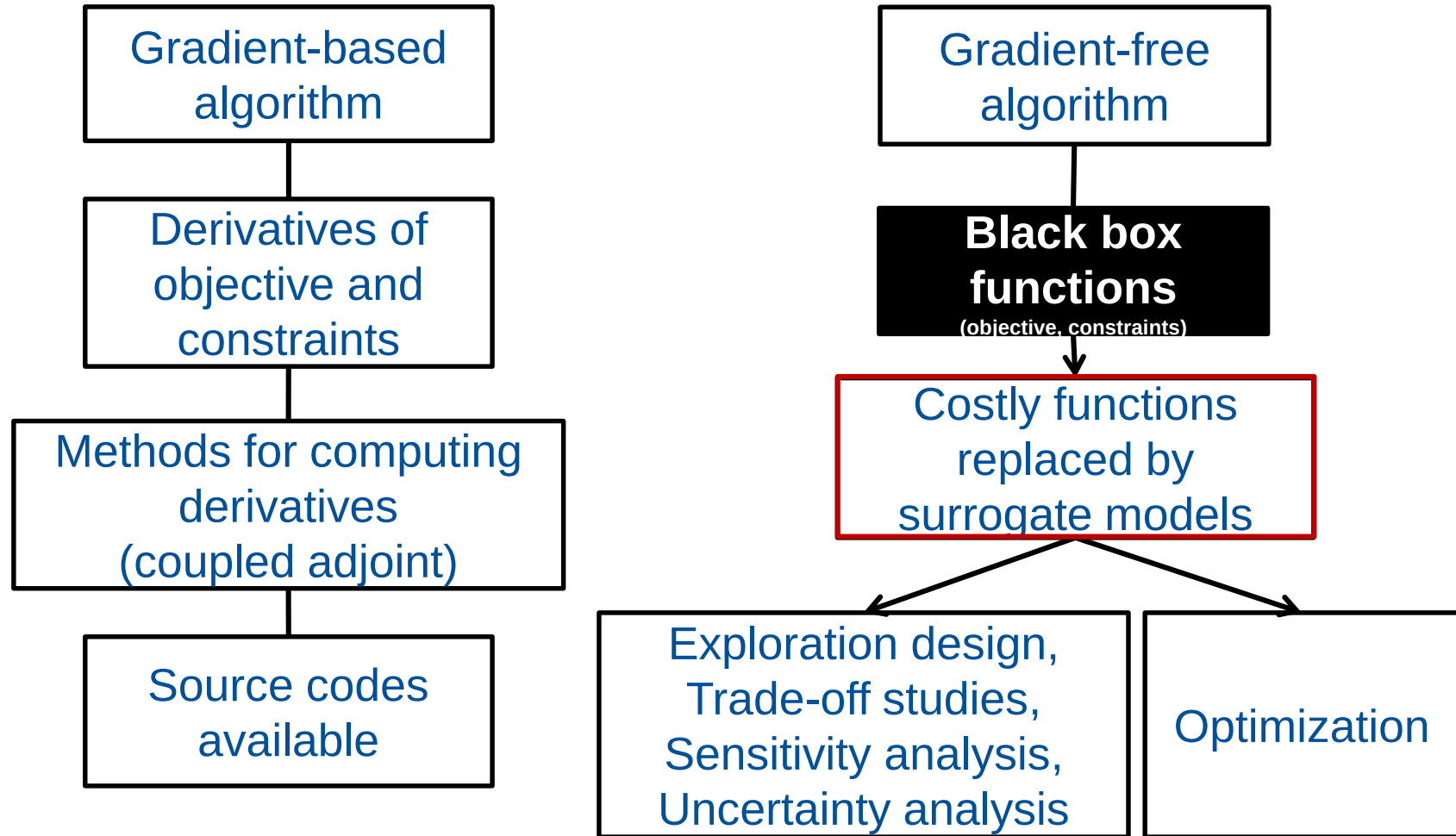


MDOLab strategy within OpenMDAO

Python user script Setup up the problem: objective function, constraints, design variables, optimizer and solver options			
Optimizer interface <i>pyOptSparse</i> Common interface to various optimization software		Aerostructural solver <i>AeroStruct</i> Coupled solution methods and coupled derivative evaluation	
Geometry modeler <i>DVGeometry/GeoMACH</i> Defines and manipulates geometry, evaluates derivatives			
SNOPT	Other optimizers	Flow solver <i>ADflow</i> Governing and adjoint equations	Structural solver <i>TACS</i> Governing and adjoint equations

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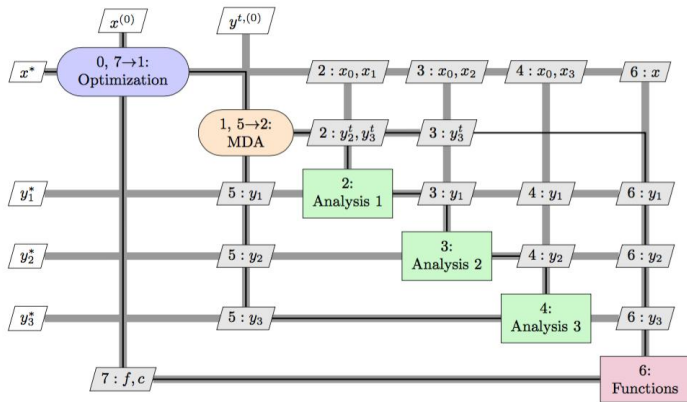


Audet, C., & Hare, W. (2017). Derivative-free and blackbox optimization. Berlin: Springer International Publishing

Powell, M. J. (1994). A direct search optimization method that models the objective and constraint functions by linear interpolation. In Advances in optimization and numerical analysis (pp. 51-67). Springer, Dordrecht.

Forrester, A., Sobester, A., & Keane, A. (2008). Engineering design via surrogate modelling: a practical guide. John Wiley & Sons

Different strategies



**Generic approach
Extension to
multiple disciplines
& mixed variables**

Gradient-free
algorithm

**Black box
functions**
(objective, constraints)

Costly functions
replaced by
surrogate models

Exploration design,
Trade-off studies,
Sensitivity analysis,
Uncertainty analysis

Optimization

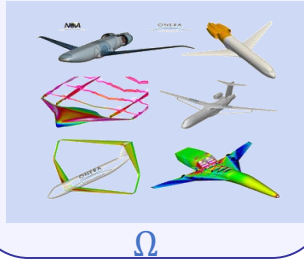
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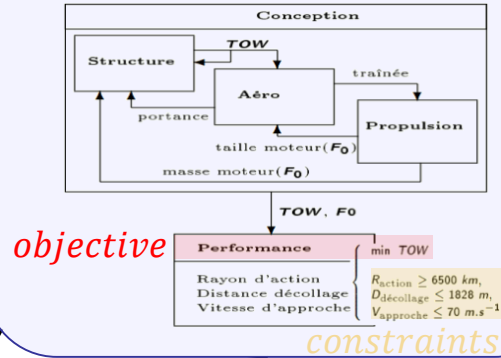
Overview

New concepts



Analyze

Multidisciplinary design analysis



Improvements

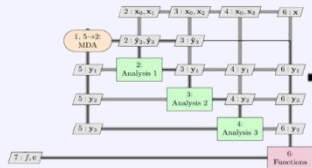
Global optimization

$$\begin{aligned} \min_{\text{inputs} \in \Omega} & \text{objective (inputs)} \\ \text{s.t.} & \text{constraints (inputs)} \end{aligned}$$

Expensive computations

Expensive black-box optimization

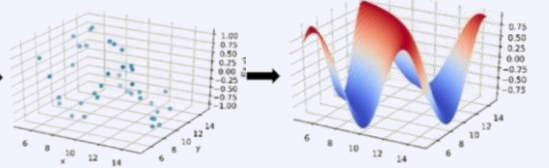
Surrogate model



Expensive black-box

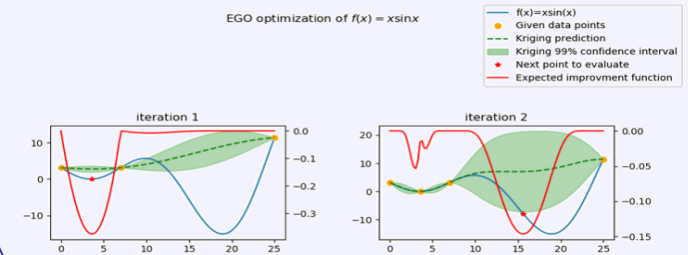
Design of experiments

Gaussian processes



Bayesian optimization

EGO optimization of $f(x) = x \sin x$

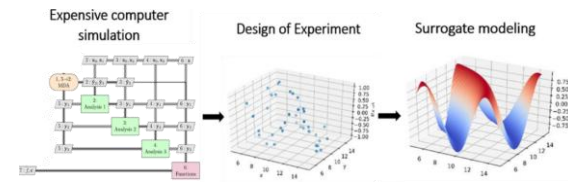


Outline

- Kriging based surrogate models
- Bayesian optimization
 - mono & multiobjective
- Applications
 - DRAGON (hybrid electric aircraft)
 - EU project AGILE4.0 & COLOSSUS

& dedicated Notebook for analytical functions

Main steps to build a surrogate model



1. Black box to replace by a surrogate model

$$y(x^{(i)}) = f(x^{(i)}) \text{ with } x^{(i)} \in \mathbb{R}^d \text{ and } y(x^{(i)}) \in \mathbb{R}$$

where f is the model (expensive, free-derivative, noise-free) such as CFD code or FEM code

2. Data (X,y): n observation points – design of experiments

- DOE** {
- Design variables (inputs)

$$\mathbf{X}^{DOE} = \{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}^t \text{ with } x^{(i)} \in \mathbb{R}^d$$
 - Observations (output responses of your expensive black box)

$$\mathbf{y}^{DOE} = \{y(x^{(1)}), y(x^{(2)}), \dots, y(x^{(n)})\}^t \text{ with } y(x^{(i)}) \in \mathbb{R}$$

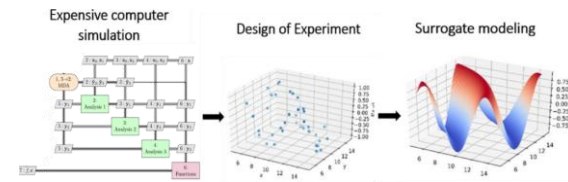
3. Fit the surrogate model: determine parameters (training)

4. Predict the response at a new point $x_{new} \in \mathbb{R}^d$

5. Validate the surrogate

$$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y(x^{(i)}))^2}{n}}$$

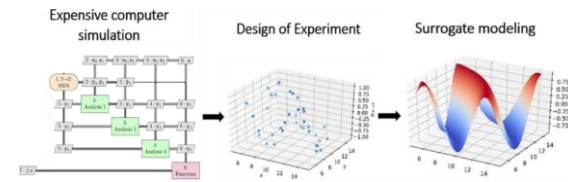
Design Of Experiments (DOE)



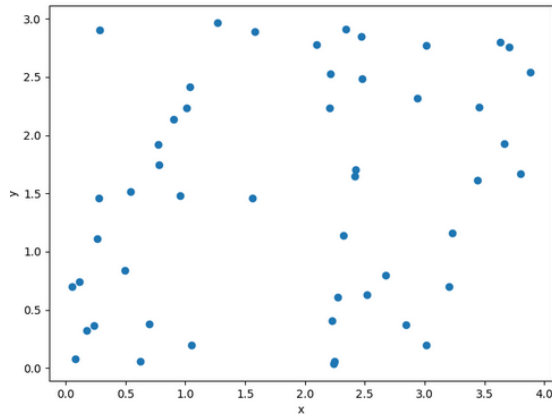
- Trade-off /exploration
 - rule of thumb with number of DOE points $> 10 \times$ number of design variables (more in high dimension)
 - Space filling designs: Full factorial or Latin Hypercube Sampling
- Sensitivity Analysis (determination of variables influent on the QoI)
 - Local sensitivity: partial derivative information to build
 - Global sensitivity: screening method (local derivatives with Morris approach) or sensitivity indices (Sobol' indices) based on MC sampling or Metamodel (Polynomial Chaos Expansion)
- Optimization
 - adaptive DOE: start with a small number of points ($3 \times$ number of design variables) and add points in interesting area until the computational budget is reached

Build a Design of Experiments (DOE)

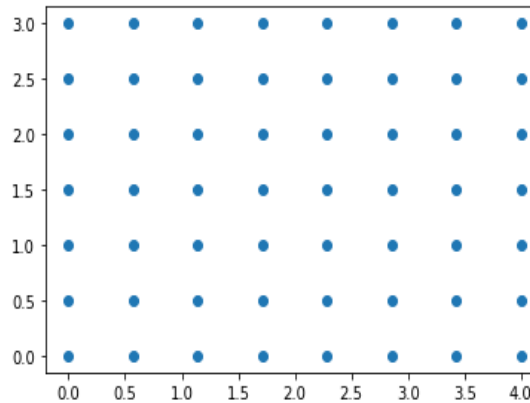
to choose N points $x \in \mathbb{R}^d$



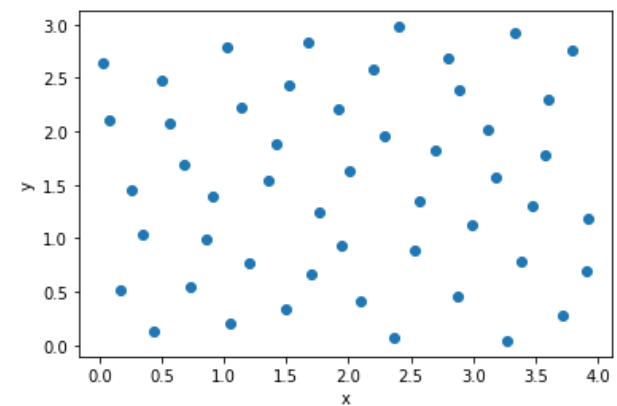
Random sampling



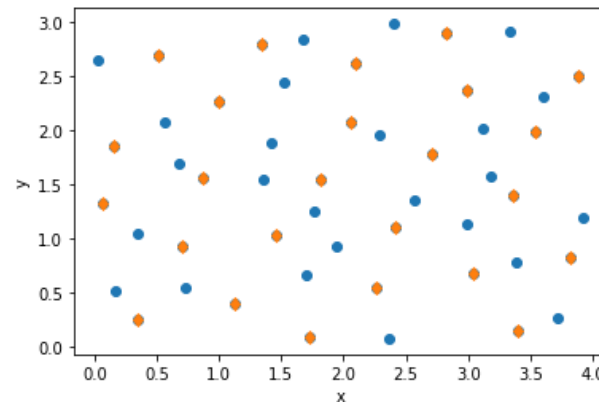
Full factorial sampling



Latin hypercube sampling (LHS)



Nested Latin hypercube sampling



- Multi-fidelity
- Validation/training
- ...

Methodology developments: Surrogate Models

Definition of a metamodel library dedicated to Aircraft design

- Models to handle a **large number of design variables**
 - ➔ New Kriging models: KPLS & KPLS-K
- Models to handle **heterogeneous functions**
 - ➔ Mixture of experts (MOE)
- Models to handle **heterogeneous variables**
 - ➔ Kriging based on continuous relaxation
- Models to handle multifidelity data
 - ➔ Co-Kriging (MFKPLS, MFKPLS-K)
 - ➔ Co-Kriging with heteroscedastic Noise



SMT 2.15 features (Sept 2026):

github.com/SMTorg/smt

- Models to handle a large number of design variables (KPLS – KPLSK – MGP)
- Mixture of experts to handle heterogeneous functions (MOE)
- Different covariance kernels added
- Multi-fidelity models (MFK – MFKPLS – MFKPLSK)
- Noisy kriging to handle uncertainties on data
- Kriging models for mixed variables (continuous, discrete, categorical) & associated kernels
- Sparse GP models to handle large database (mono and multifidelity)
- Bayesian optimization (EGO without constraint) for continuous and mixed variables

➔ Included some Jupyter notebooks



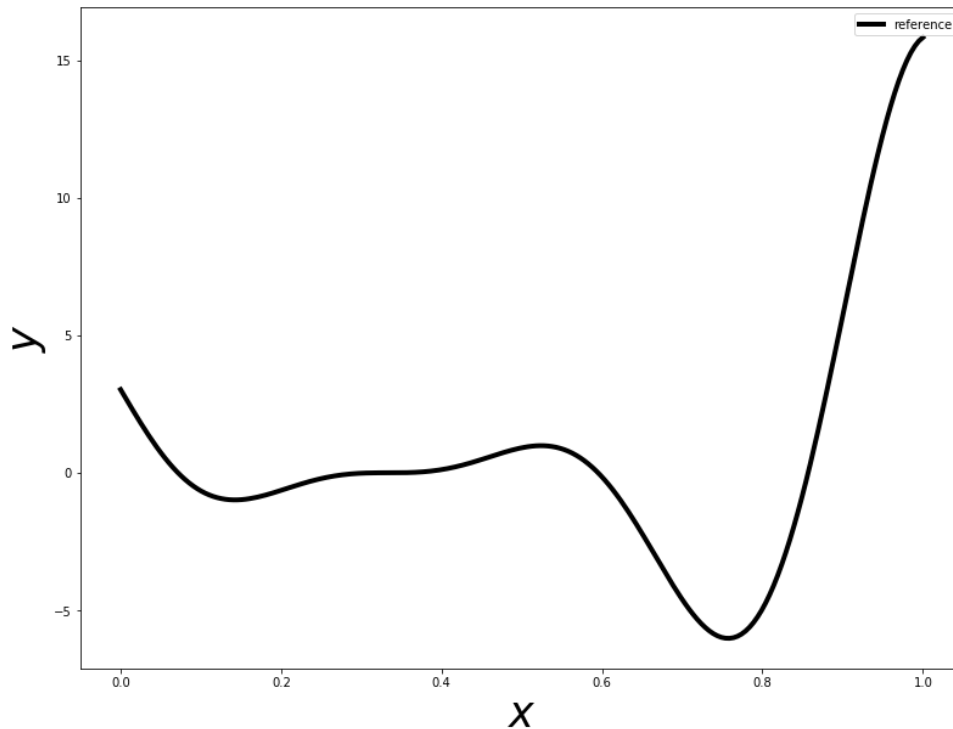
Bouhlef, M. A., Hwang, J. T., Bartoli, N., Lafage, R., Morlier, J., & Martins, J. R. (2019). A Python surrogate modeling framework with derivatives. *Advances in Engineering Software*, 135, 102662.

Bouhlef, M. A., Bartoli, N., Otsmane, A., and Morlier, J., "Improving kriging surrogates of high-dimensional design models by Partial Least Squares dimension reduction," *Structural and Multidisciplinary Optimization*, Vol. 53, No. 5, 2016, pp. 935–952.

Bouhlef, M. A., Bartoli, N., Otsmane, A., and Morlier, J., "An Improved Approach for Estimating the hyperparameters of the Kriging Model for high-dimensional problems through The Partial Least Squares Method", *Mathematical Problems in Engineering*, Vol. 2016(4), May 2016

Gaussian process or Kriging model

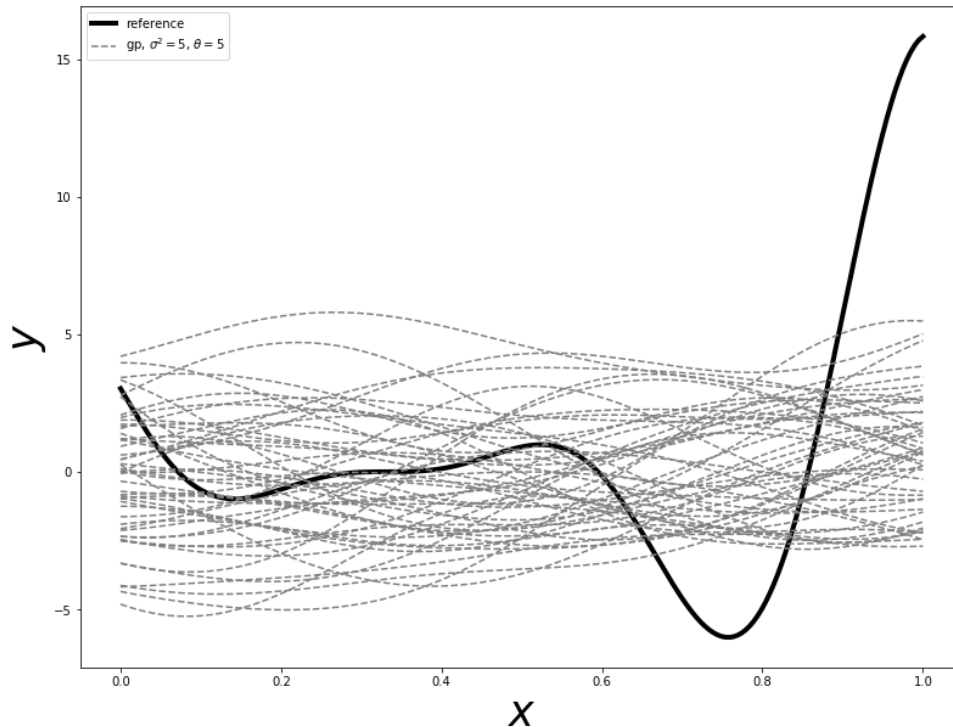
$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$



D. G. Krige. A statistical approach to some basic mine valuation problems on the witwatersrand. Journal of the Southern African Institute of Mining and Metallurgy, 52(6):119–139, 1951
[C. E. Rasmussen and C. K. Williams. Gaussian processes for machine learning. volume 1. MIT press Cambridge, 2006.](#)

Gaussian process or Kriging model

$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$



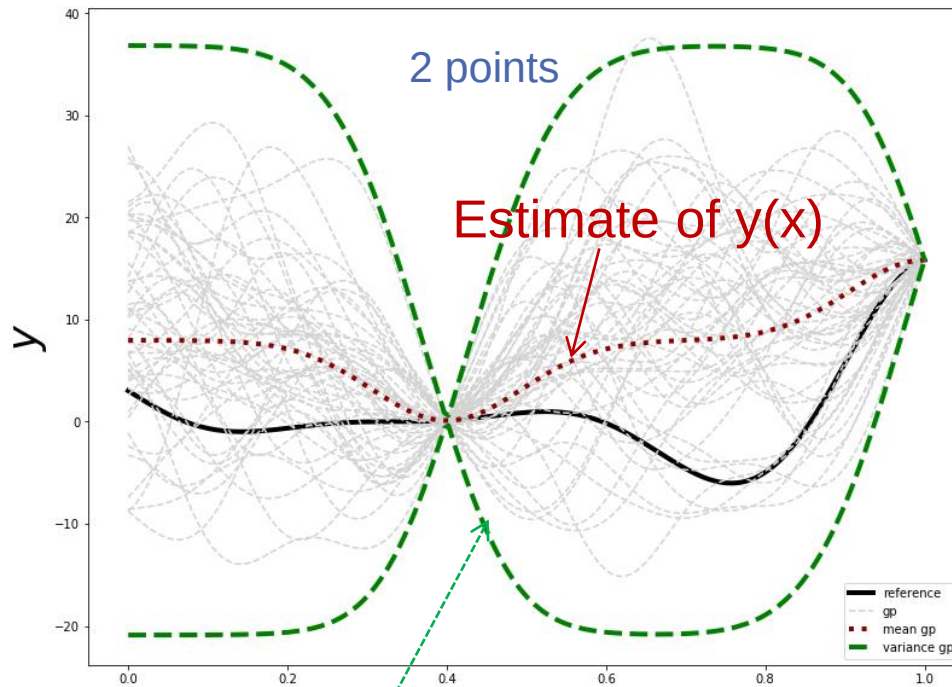
Gaussian process
characterized by:

- its mean (or trend)
 $\mu(x) \in \mathbb{R}$
- its covariance Kernel
 $k(x, x') \in \mathbb{R}$

D. G. Krige. A statistical approach to some basic mine valuation problems on the witwatersrand. Journal of the Southern African Institute of Mining and Metallurgy, 52(6):119–139, 1951
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Gaussian process or Kriging model

$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$



Gaussian process characterized by:

- its mean (or trend)

$$\mu(x) \in \mathbb{R}$$

- its covariance Kernel

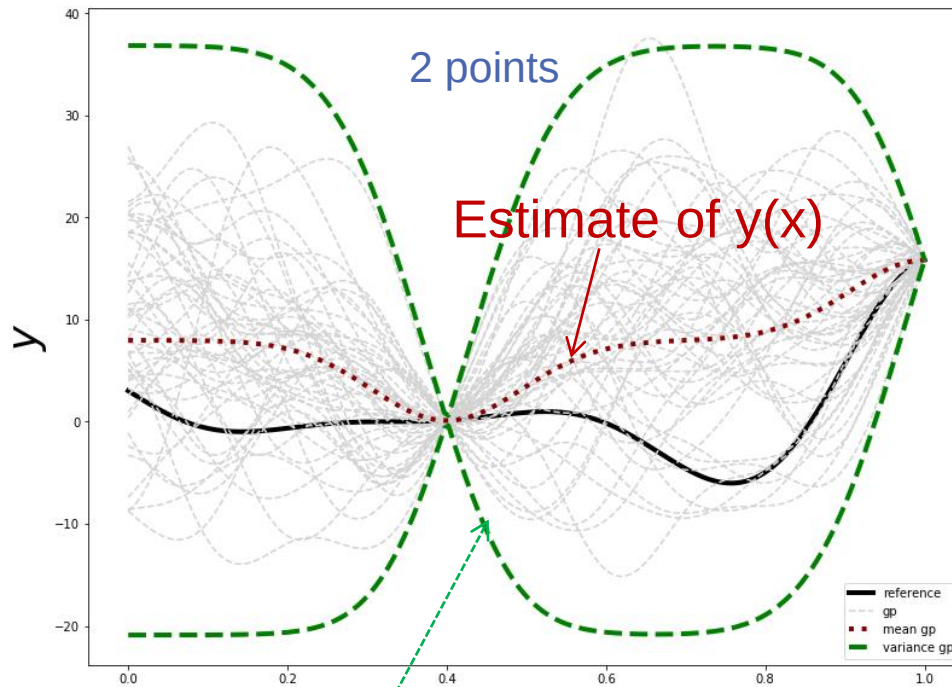
$$k(x, x') \in \mathbb{R}$$

+ information provided by data

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Gaussian process or Kriging model

$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$



Gaussian process
characterized by:

- its mean (or trend)
- its covariance Kernel

$$\mu(x) \in \mathbb{R}$$

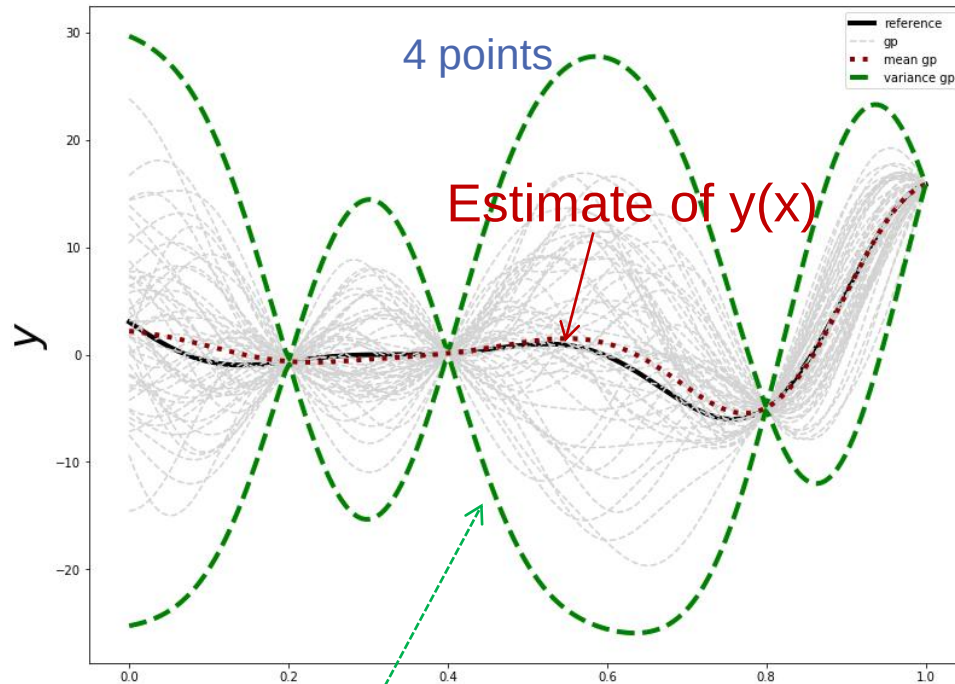
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Gaussian process or Kriging model

$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$



Quantification of the uncertainties in these estimates

Gaussian process characterized by:

- its mean (or trend)

$$\mu(x) \in \mathbb{R}$$

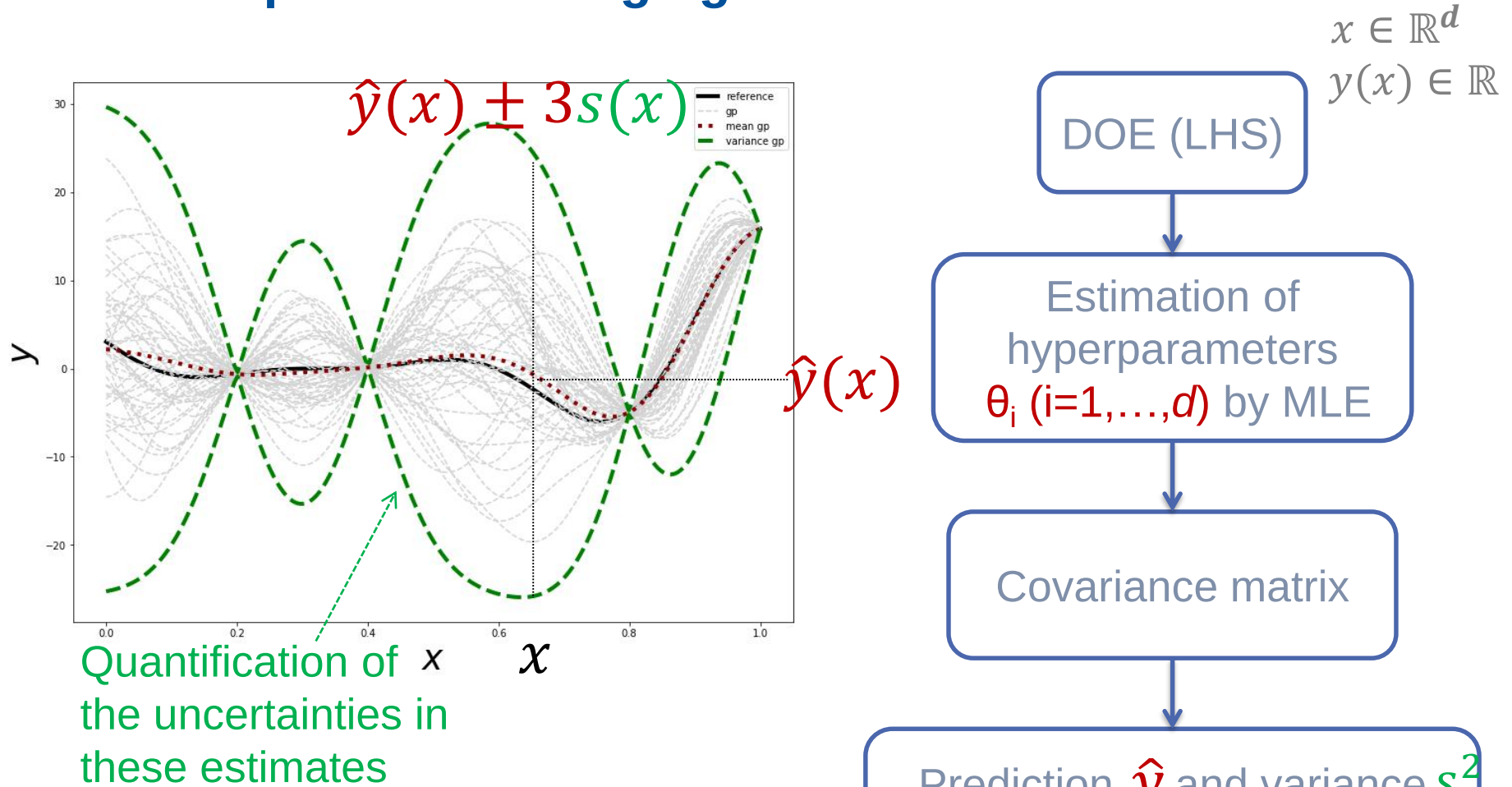
- its covariance Kernel

$$k(x, x') \in \mathbb{R}$$

+ information provided by data

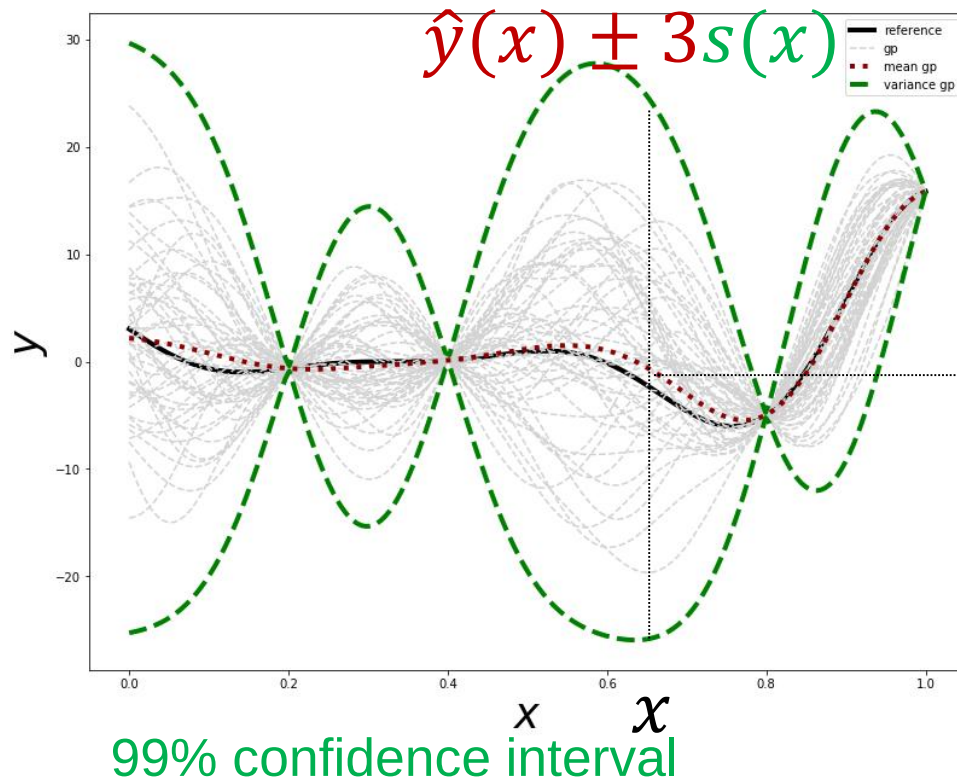
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Gaussian process or Kriging model



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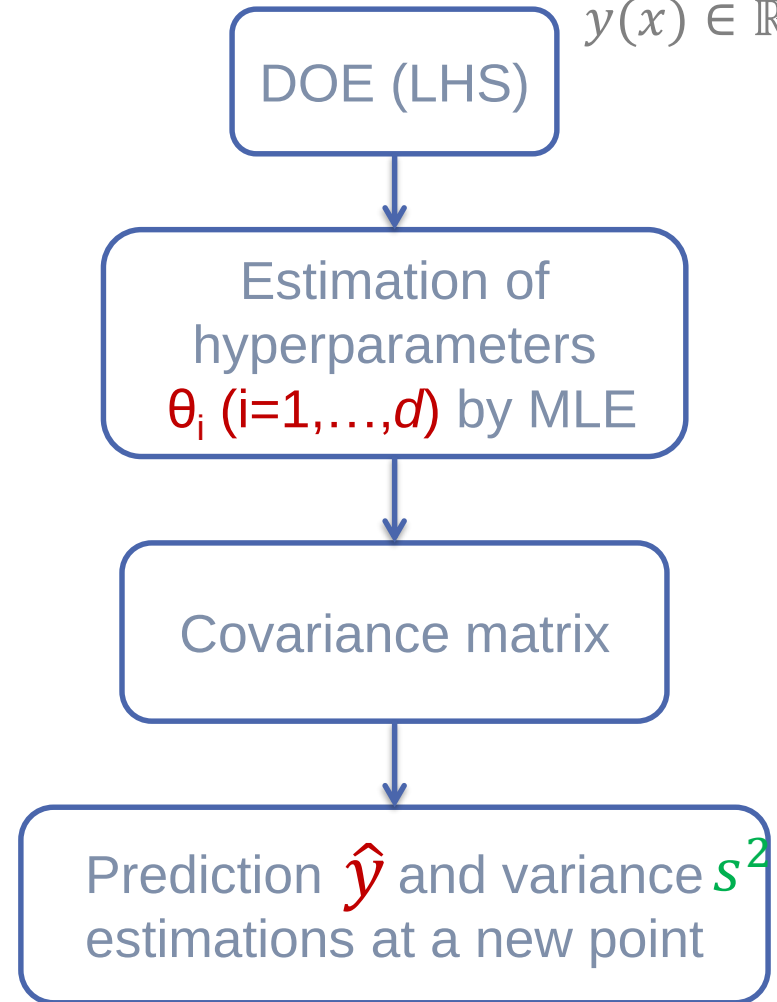
Gaussian process or Kriging model



$$f(x) \Rightarrow Y(x) = \mathcal{N}(\hat{y}(x), s^2(x))$$

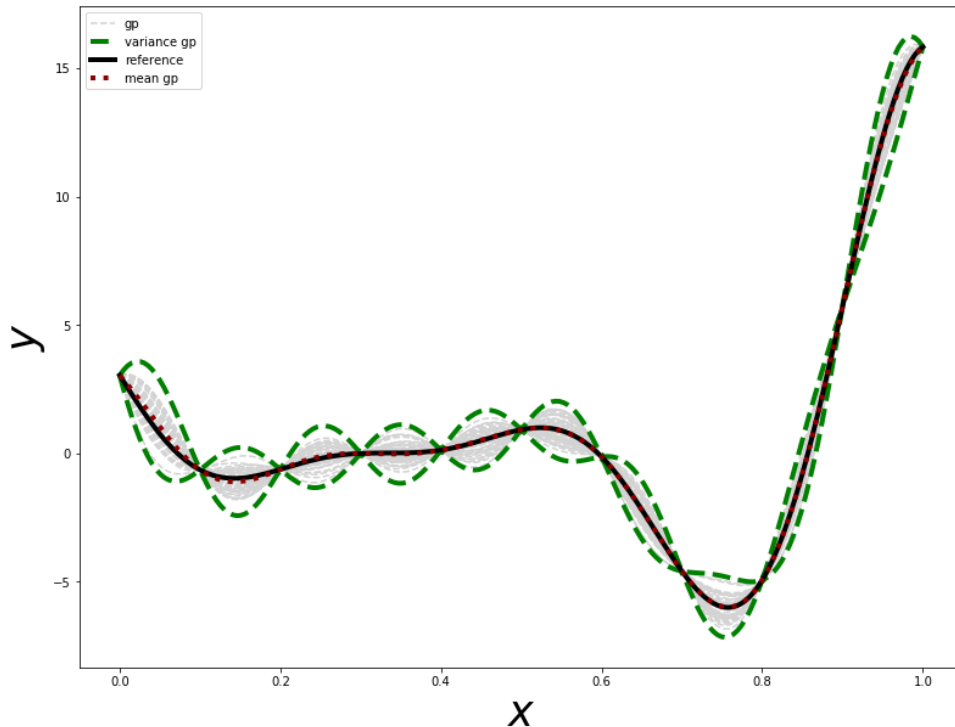
$$x \in \mathbb{R}^d$$

$$y(x) \in \mathbb{R}$$



D. G. Krige. A statistical approach to some basic mine valuation problems on the witwatersrand. Journal of the Southern African Institute of Mining and Metallurgy, 52(6):119–139, 1951
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Gaussian process or Kriging model



- ➔ Hyperparameters tuning
- ➔ Number of hyperparameters increases with the dimension d (number of design variables)
- ➔ Curse of dimensionality

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DOE (LHS)

Estimation of
hyperparameters
 θ_i ($i=1,\dots,d$) by MLE

Covariance matrix

Prediction $\hat{y}$ and variance s^2
estimations at a new point

$x \in \mathbb{R}^d$
 $y(x) \in \mathbb{R}$

Hyperparameters tuning

Given observed training data: $\mathbf{Y} = \{y_1 \dots y_N\}$

$$f(x) \Rightarrow Y(x) \sim \mathcal{N}(\textcolor{red}{m}(X), \textcolor{green}{K}(\theta) + \sigma^2(x))$$

Maximum Likelihood Estimation (MLE): find the parameters (θ, σ) that maximize the probability of observing this data.

The marginal log-likelihood objective:

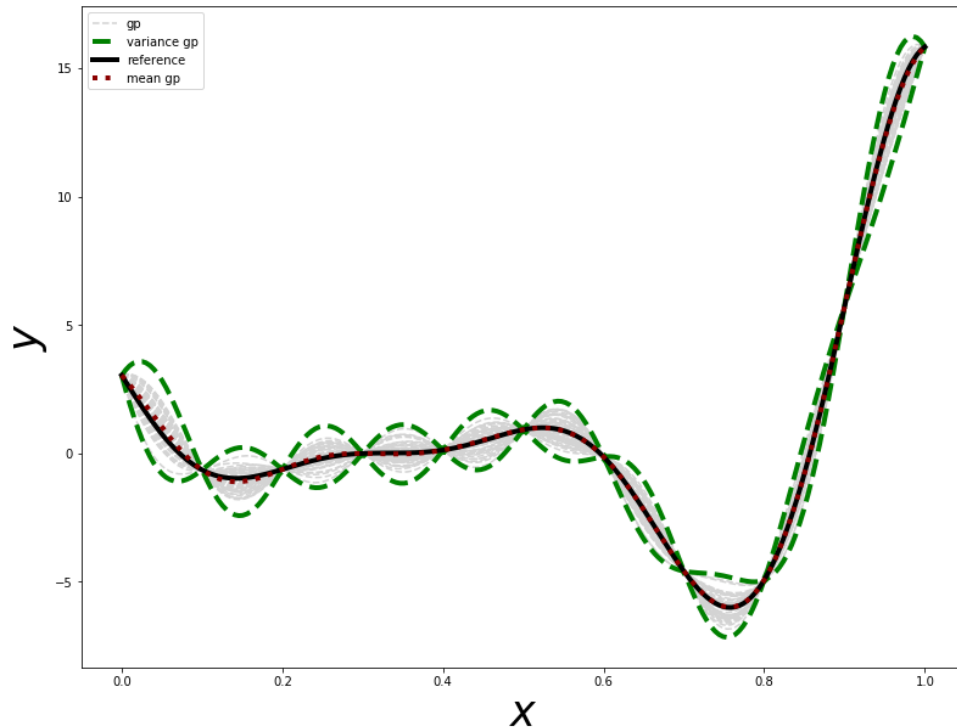
$$-\log p(Y|X, \theta, \sigma) = \frac{1}{2} \frac{Y^T K^{-1} Y}{\sigma^2} + \frac{1}{2} \log(|K|) + \frac{N}{2} \log(2\pi) + \frac{N}{2} \log(\sigma^2)$$

➔ Use gradient based algorithms to compute the hyperparameters (θ, σ)

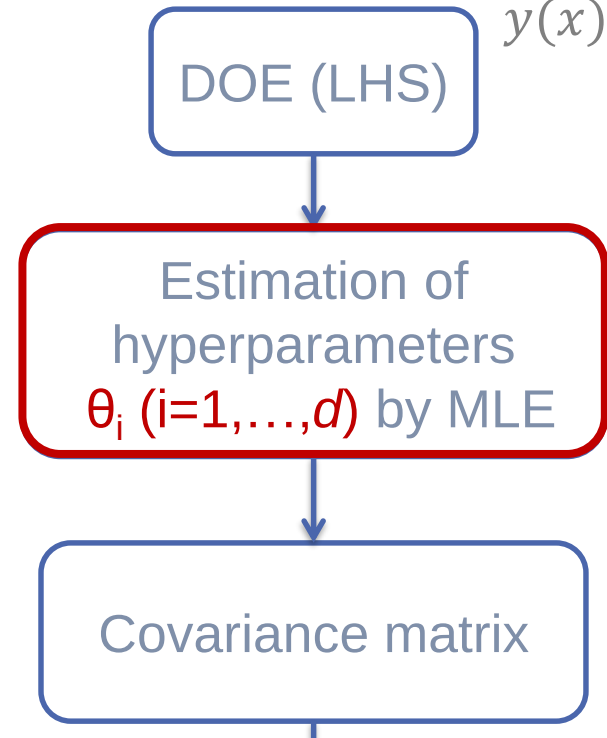
Exemple of the squared exponential kernel

$$k(x, x') = \sigma^2 \exp \left(- \sum_{i=1}^d \textcolor{red}{\theta}_i |x_i - x'_i|^2 \right)$$

Gaussian process or Kriging model



$$x \in \mathbb{R}^d$$
$$y(x) \in \mathbb{R}$$

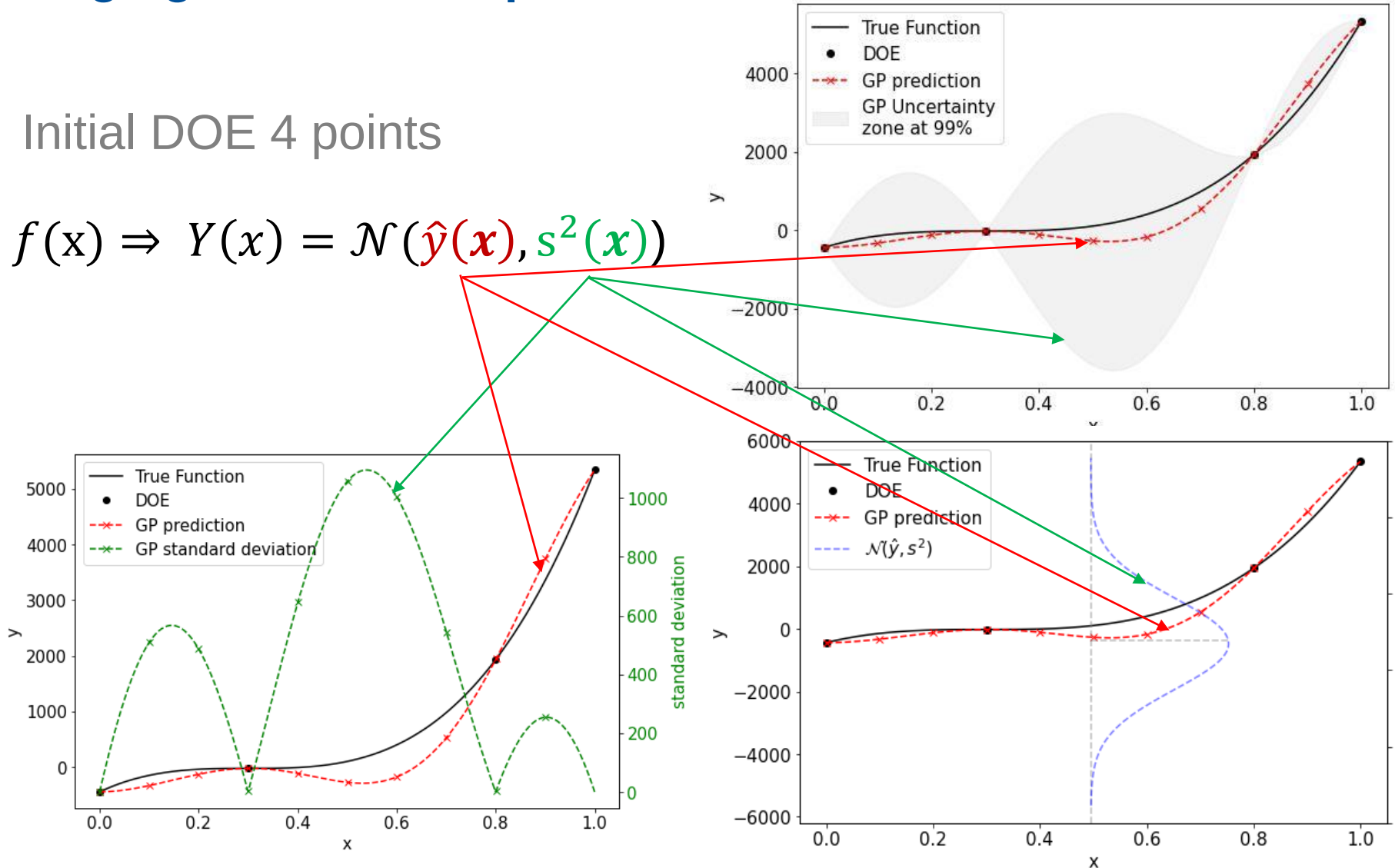


GP approach in the notebook

Kriging or Gaussian process model

Initial DOE 4 points

$$f(x) \Rightarrow Y(x) = \mathcal{N}(\hat{y}(x), s^2(x))$$



D. G. Krige. A statistical approach to some basic mine valuation problems on the witwatersrand. Journal of the Southern African Institute of Mining and Metallurgy, 52(6):119–139, 1951
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Models to handle a large number of design variables

New Kriging models: KPLS & KPLS-K

➔ Exploitation of information provided by PLS (Partial Least Squares) in the construction of the Kriging model to reduce the dimension: KPLS and KPLS-K models

$$x \in \mathbb{R}^d$$

Ordinary
Kriging

$$k(x, x') = \sigma^2 \exp \left(- \sum_{i=1}^d \theta_i |x_i - x'_i|^{p_i} \right) \quad \text{with}$$

d parameters θ_i
to evaluate

Covariance kernel



KPLS

$$k_{PLS}(x, x') = \sigma^2 \exp \left(- \sum_{i=1}^d \eta_i |x_i - x'_i|^{p_i} \right) \quad \text{with}$$

$$\eta_i = \sum_{j=1}^h \theta_j |w_{i,j}|^{p_i}$$

h parameters θ_j to evaluate

- $|w_{i,j}|_{i=1,\dots,d}$ describes how sensitive the j -th principal component is to each design variable i ➔ PLS
- θ_j describes how sensitive the function is to each principal component (max $h \approx 4$) ➔ MLE
- If $h = d$ ➔ classical Kriging (exponential kernels)

Wold H (1966) Estimation of Principal Components and Related Models by Iterative Least squares, Academic Press, New York, pp 391–420

Bouhlef, M. A., Bartoli, N., Otsmane, A., and Morlier, J., "Improving kriging surrogates of high-dimensional design models by Partial Least Squares dimension reduction," Structural and Multidisciplinary Optimization, Vol. 53, No. 5, 2016, pp. 935–952.

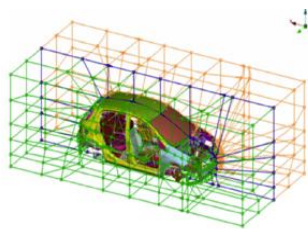
Bouhlef, M. A., Bartoli, N., Otsmane, A., and Morlier, J., "An Improved Approach for Estimating the hyperparameters of the Kriging Model for high-dimensional problems through The Partial Least Squares Method", Mathematical Problems in Engineering, Vol. 2016(4), May 2016

Models to handle a large number of design variables

New Kriging models: KPLS & KPLS-K

MOPTA test case function
from automotive industry

d=124 inputs 1 output
training: 500 points LHS,
validation: 100 points



$$RE = \frac{\|y - \hat{y}\|_2}{\|y\|_2} 100$$

SNECMA test case
(turbomachinery)

d=98 inputs 1 output
training: 340 points LHS,
validation: 24 points LHS



Surrogate	RE (%)	CPU time
Ordinary Kriging (Scikit-Learn)	3.28e-7	17 min 23 s
		Time / 28
KPLS $h=4$	4.52e-7	37 s

Surrogate	RE (%)	CPU time
Ordinary Kriging (Snecma ref)	2.24	1min 33s
		Time / 60
KPLS $h=1$	1.62	0.90 s
KPLS $h=2$	1.62	1.56 s

Intel(R) Core(TM) i7-4500U CPU@1.80GHz, 6.00 Go RAM

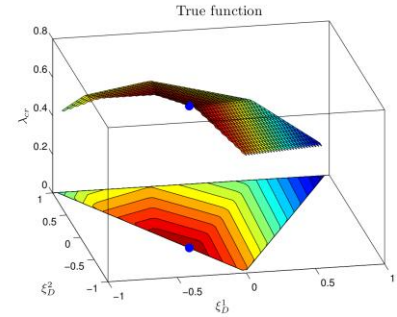
Intel(R) Xeron(R) CPU W3565@3.20GHz, 7.98 Go RAM
Quad core

- CPU time drastically reduced: interest for adaptive enrichment optimization method
- Automatic choice for the number of PLS components

Jones, D., "Large-scale multi-disciplinary mass optimization in the auto industry," MOPTA 2008 Conference (20 August 2008)
Bouhlel, M.-A., Ph.D. thesis, ISAE-SUPAERO, 2016. <https://hal.archives-ouvertes.fr/tel-01293319>

Models to handle heterogeneous functions

Mixture of Experts (MOE)



→ Mixture of experts technique

- Divide the database into K clusters (Expectation-Maximization)
- Build a local surrogate model on each cluster (RBF, Polynomial functions, Kriging,...)
- Recombine the K local models into a global model

$$\hat{f}(x) = \sum_{i=1}^K P(k = i/X = x) \hat{f}_i$$

K number of clusters (Gaussian components)

$P(k = i/X = x)$ probability to be in the cluster i

(posterior probability given by the Expectation-Maximization algorithm)

$\hat{f}_i$ local expert build using the points in cluster i (RBF, Polynomial functions, Kriging,...)

Jordan, M. I., Jacobs, R. A., "Hierarchical mixtures of experts and the EM algorithm", Neural Comput. 6 (1994) 181–214.

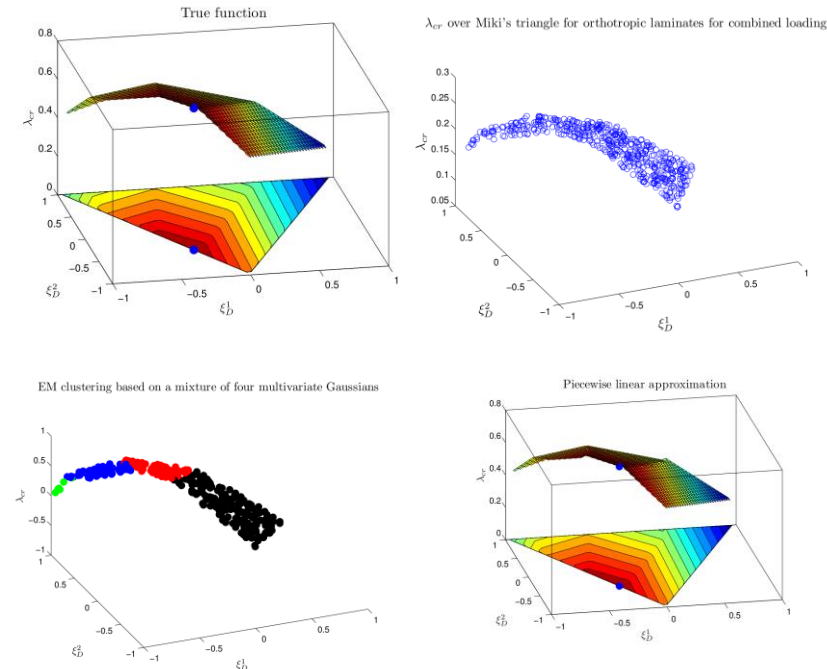
Bettebghor, D., Bartoli, N., Grihon, S., Morlier, J., and Samuelides, M., "Surrogate modeling approximation using a mixture of experts based on EM joint estimation," Structural and Multidisciplinary Optimization, Vol. 43, No. 2, 2011, pp. 243–259

Liem, R. P., Mader, C. A., and Martins, J. R. R. A., "Surrogate Models and Mixtures of Experts in Aerodynamic Performance Prediction for Mission Analysis," Aerospace Science and Technology, Vol. 43, 2015, pp. 126–151

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Comparison on Buckling critical loads PhD D. Bettebghor 2011

Jordan, M. I., Jacobs, R. A., "Hierarchical mixtures of experts and the EM algorithm", Neural Comput. 6 (1994) 181–214.

Bettebghor, D., Bartoli, N., Grihon, S., Morlier, J., and Samuelides, M., "Surrogate modeling approximation using a mixture of experts based on EM joint estimation," Structural and Multidisciplinary Optimization, Vol. 43, No. 2, 2011, pp. 243–259

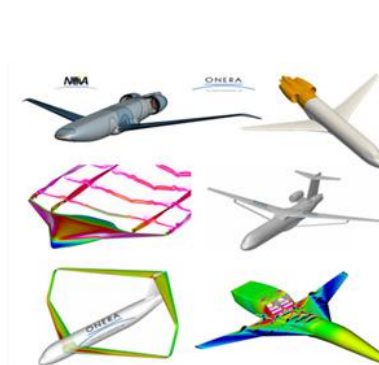
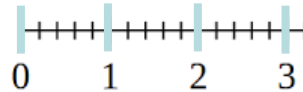
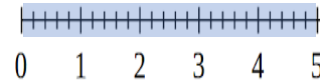
Liem, R. P., Mader, C. A., and Martins, J. R. R. A., "Surrogate Models and Mixtures of Experts in Aerodynamic Performance Prediction for Mission Analysis," Aerospace Science and Technology, Vol. 43, 2015, pp. 126–151

Models to handle mixed variables (continuous, discrete, categorical)

Hybrid variables

Variables types:

- **Continuous (x)** Ex: wing length
- **Integer (z)** Ex: winglet number
- **Categorical (u)** Ex: Plane shape / material properties



Categorical variables: n variables,
 $n=2$

$u1$ = shape

$u2$ = color

Levels: L_i levels for i in $1, \dots, n$,
 $L_1=3, L_2=2$

Levels($u1$)= square, circle, rhombus

Levels($u2$)= blue, red

Categories: $\prod_{i=1}^n L_i, 2*3=6$

- Blue square
- Blue circle
- Blue rhombus
- Red square
- Red circle
- Red rhombus

6 possibilities

State of the Art approach: Continuous Relaxation

Ex: Garrido-Merchán and Hernández-Lobato model

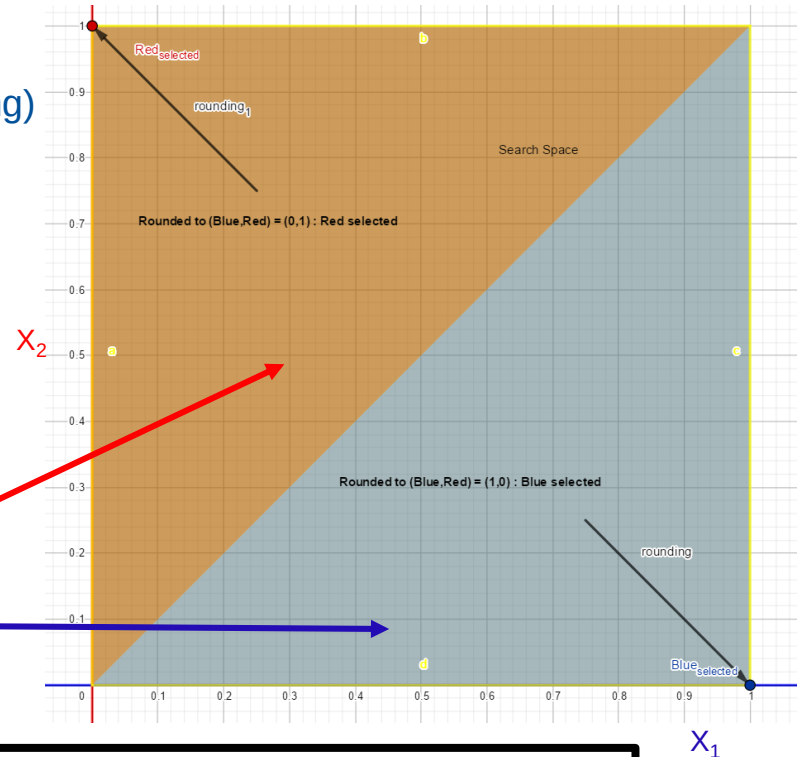
➔ Model as a **Continuous Relaxation** (one-hot-encoding)

Example with 1 categorical variable X and two levels

- Red color
- Blue color

➔ 1 Categorical variable replaced by 2 continuous variables denoted by X_1 and $X_2 \in [0,1]$

- If $X_1 > X_2 \Rightarrow (1., 0.) \Rightarrow$ Blue color
- If $X_1 < X_2 \Rightarrow (0., 1.) \Rightarrow$ Red color



A continuous kernel

(in the relaxed dimension)

$$k(x^r, x^s, \theta^{cont}) = \prod_{j=1}^{d'} \exp \left(-(x_j^r - x_j^s) \theta_j^{cont} (x_j^r - x_j^s) \right)$$

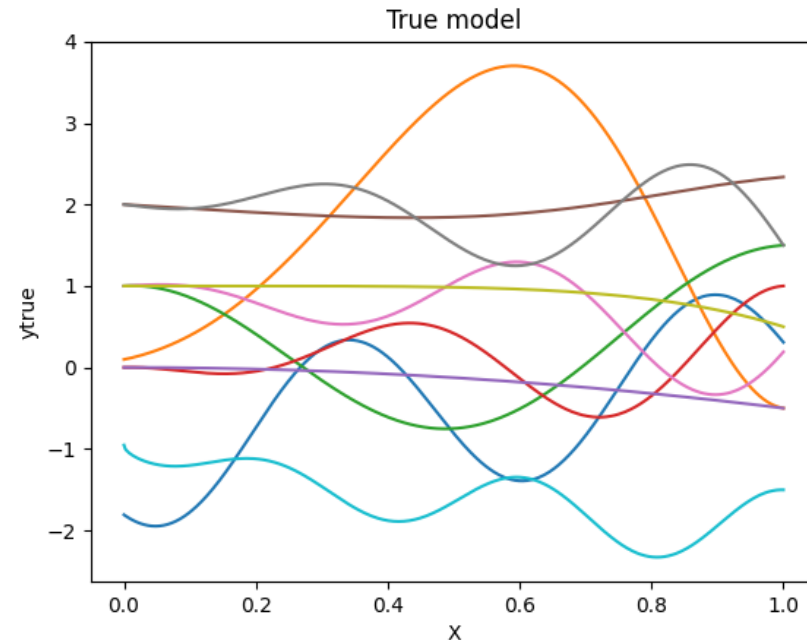
➔ Use of KPLE models to decrease the dimension

E. C. Garrido-Merchán, and D. Hernández-Lobato. "Dealing with categorical and integer-valued variables in Bayesian Optimization with Gaussian processes". Neurocomputing, vol. 380 (2020), pages 20-35
Saves, P., Bartoli, N., Diouane, Y., Morlier, J., & Lefebvre, T. (2021, March). Enhanced Kriging models within a Bayesian optimization framework to handle both continuous and categorical inputs. In SIAM CSE21

Models to handle mixed variables (continuous, discrete, categorical)

Toy function with a categorical variable (10 levels)

$$f(x, z) = \begin{cases} \cos(3.6\pi(x-2)) + x - 1 & \text{if } z = 1, \\ 2 \cos(1.1\pi \exp(x)) - \frac{x}{2} + 2 & \text{if } z = 2, \\ \cos(2\pi x) + \frac{1}{2}x & \text{if } z = 3, \\ x \left(\cos(3.4\pi(x-1)) - \frac{x-1}{2} \right) & \text{if } z = 4, \\ -\frac{x^2}{2} & \text{if } z = 5, \\ 2 \cos\left(\frac{\pi}{4} \exp(-x^4)\right) - \frac{x}{2} + 1 & \text{if } z = 6, \\ x \cos(3.4\pi x) - \frac{x}{2} + 1 & \text{if } z = 7, \\ x \left(-\cos\left(\frac{7\pi}{2}x\right) - \frac{x}{2} \right) + 2 & \text{if } z = 8, \\ -\frac{x^5}{2} + 1 & \text{if } z = 9, \\ -\cos(5\frac{\pi}{2}x)^2 \sqrt{x} - \frac{\ln(x+0.5)}{2} - 1.3 & \text{if } z = 10. \end{cases}$$



1 continuous + 1 categorical variable (10 levels)

➔ 11 continuous variables with **Continuous Relaxation** (one-hot encoding)

➔ **Partial Least Squares** techniques to reduce the number of hyperparameters

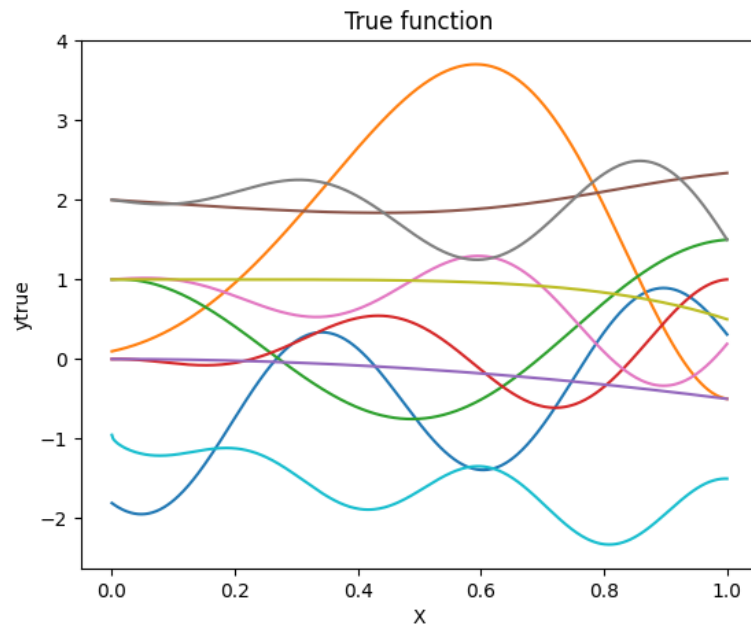
Garrido-Merchán, E. C. and Hernández-Lobato, D.. "Dealing with categorical and integer-valued variables in Bayesian Optimization with Gaussian processes". *Neurocomputing*, vol. 380 (2020), pages 20-35

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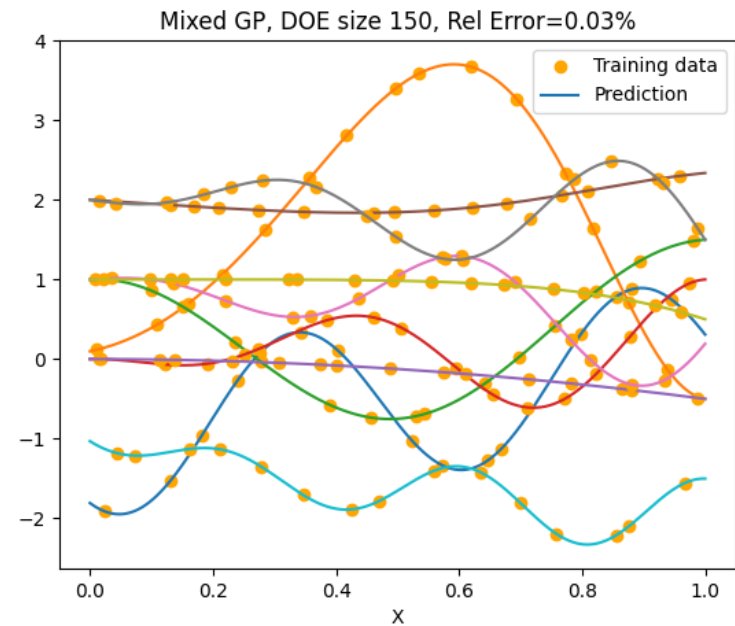
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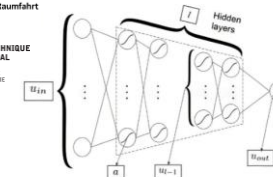
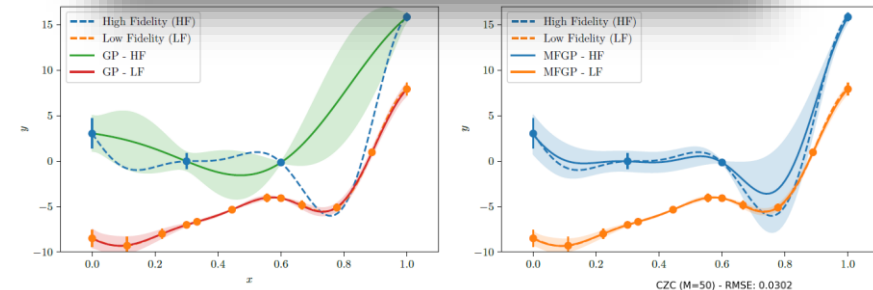
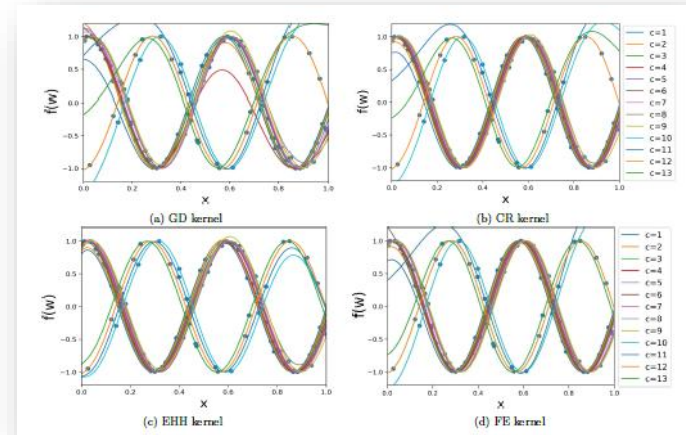
10 independant GPs



1 GP using CR

➔ Smaller errors compared to independant GPs

- **Mixed kernels integration** (Phd P. Saves)
 - Continuous Relaxation, Gower distance, Homoscedastic hypersphere, Exponential Homoscedastic hypersphere + KPLS for dimension reduction with automatic choice for number of PLS components
- **Multi-fidelity surrogates with heterogenous uncertainties on data**
 - (Phd R. Charayron – PhD M. Castano Aguirre)
- **Extension to hierarchical variables (variable-size problems)**
 - New proposed kernels (Phd P. Saves)
 - New application cases (Collab. J. Bussemaker DLR)
- **Gaussian models for large database (N points)**
 - Low-rank representation with $M < N$ inducing points
 - Training cost $O(N^3) \rightarrow O(NM^2)$

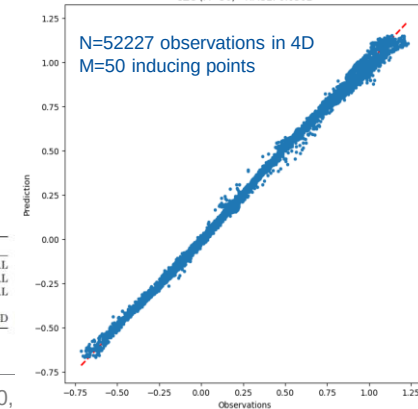


(a) Illustration of the multi-layer perceptron [2], Figure 1]

Hyperparameter of the MLP	Variable	Domain	Type	Role
Learning rate	r	$[10^{-5}, 10^{-2}]$	FLOAT	NEUTRAL
Activation function	a	{ReLU, Sigmoid}	ENUM	NEUTRAL
Batch-size	b	{8, 16, ..., 128, 256}	ORD	NEUTRAL
# of hidden layers	l	{1, 2, 3}	ORD	META
# of units hidden layer i	u_i	{50, 51, ..., 55}	ORD	DECREED

(b) Description of the variables for the problem

Figure 4: The hierarchical multi-layer perceptron problem [2].



Saves, P., Diouane, Y., Bartoli, N., Lefebvre, T., & Morlier, J. (2023). A mixed-categorical correlation kernel for Gaussian process. *Neurocomputing*, 550, 1-10.

Bussemaker, J. H., Bartoli, N., Lefebvre, T., Ciampa, P. D., & Nagel, B. (2021). Effectiveness of Surrogate-Based Optimization Algorithms for System Architecture Optimization. In *AIAA AVIATION 2021 FORUM* (p. 3095).

Valayer, H., Bartoli, N., Castañó-Aguirre, M., Lafage, R., Lefebvre, T., López-Lopera, A. F., & Mouton, S. (2024). A Python Toolbox for Data-Driven Aerodynamic Modeling Using Sparse Gaussian Processes. *Aerospace*, 11(4), 260.

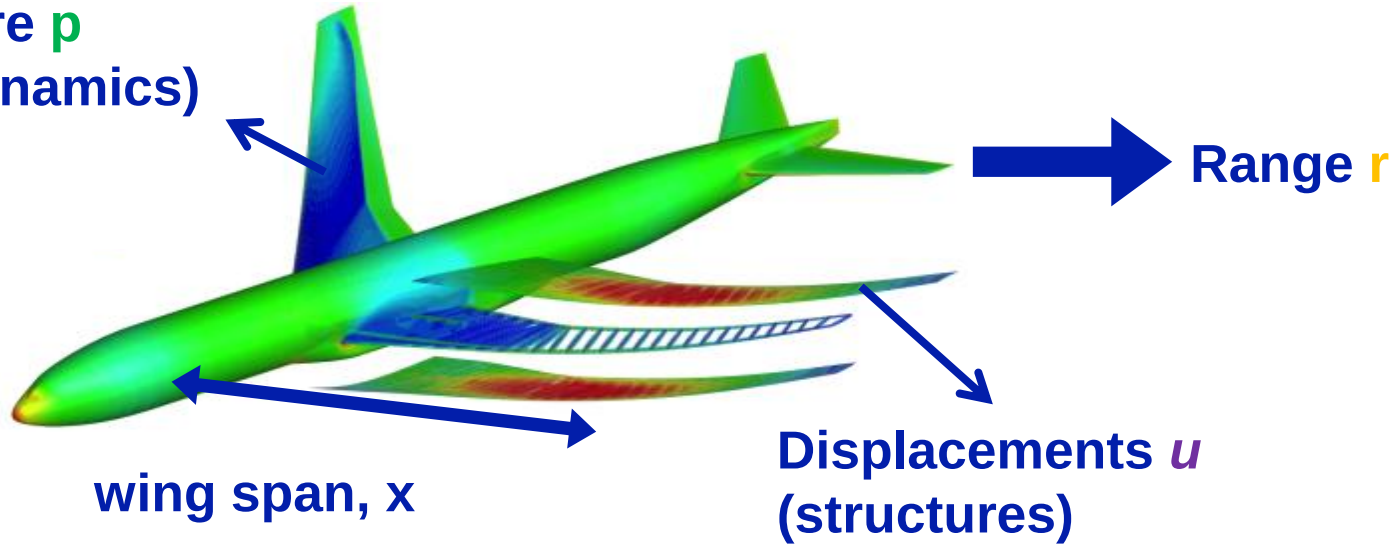
Outline

- Kriging based surrogate models
- Bayesian optimization
 - mono & multiobjective
- AGILE 4.0 & COLOSSUS applications

State of the art: Multidisciplinary Design Analysis and Optimization

Goal: Aircraft/drone design optimization

Pressure p
(aerodynamics)



min objective function $f(x, y(x))$
with respect to design variables x
subject to constraints

→ $y(x)$ are solution of a non linear system (MDA)

→ Objective and constraint functions could be costly

Joaquim R. R. A. Martins and Andrew Ning. Engineering Design Optimization. Cambridge University Press, 2020.

Optimization problem in the field of aircraft design

$$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^d} & \mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})] & 1 \text{ to } n \text{ objectives} \\ \text{s.t.} & & d \text{ design variables} \\ c_1(\mathbf{x}) \leq 0 \dots c_j(\mathbf{x}) = 0 \dots c_m(\mathbf{x}) \leq 0 & & m \text{ mixed constraints (eq. \& ineq)} \end{cases}$$

- **Main characteristics for aircraft design problem**
 - **Mono & Multi** objective, multi-constraints (1 ~ 100 constraints)
 - Intermediate dimension problem (1 ~ 100 variables)
 - Costly evaluation (CFD, FEM, objective and/or constraints)
 - Handling **non linear constraints** (black box, no derivative available)
- **Applications**
 - Disciplinary solvers (aerodynamic, structure, propulsion, ...)
 - Overall aircraft design process

How to optimize some costly functions?

- Replace the black box function (costly code, objectif or constraints) by its surrogate approximation in the optimization process
 - Build an accurate surrogate model for the costly function
 - Replace the function by its approximation in the optimization
- ➔ Build an accurate surrogate model requires lots of function evaluations
- Use an iterative process to minimize the number of function evaluations

Minimize directly the surrogate on a 1D example

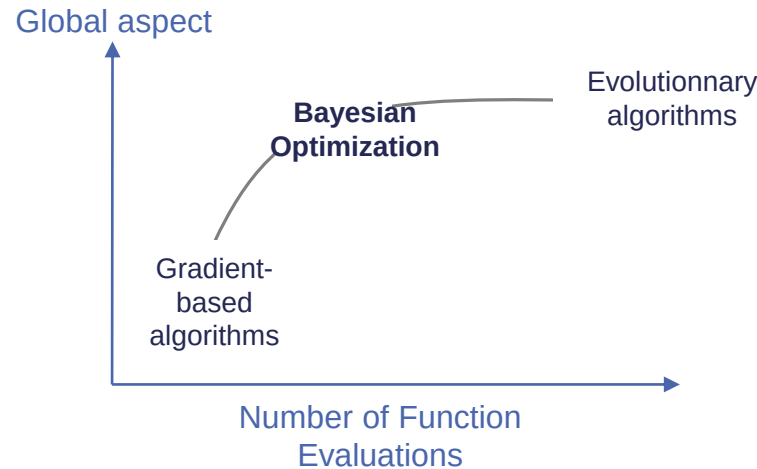
$$f(x) = (6x - 2)^2 \sin(12x - 4)$$



Surrogate Based Optim approach in the notebook

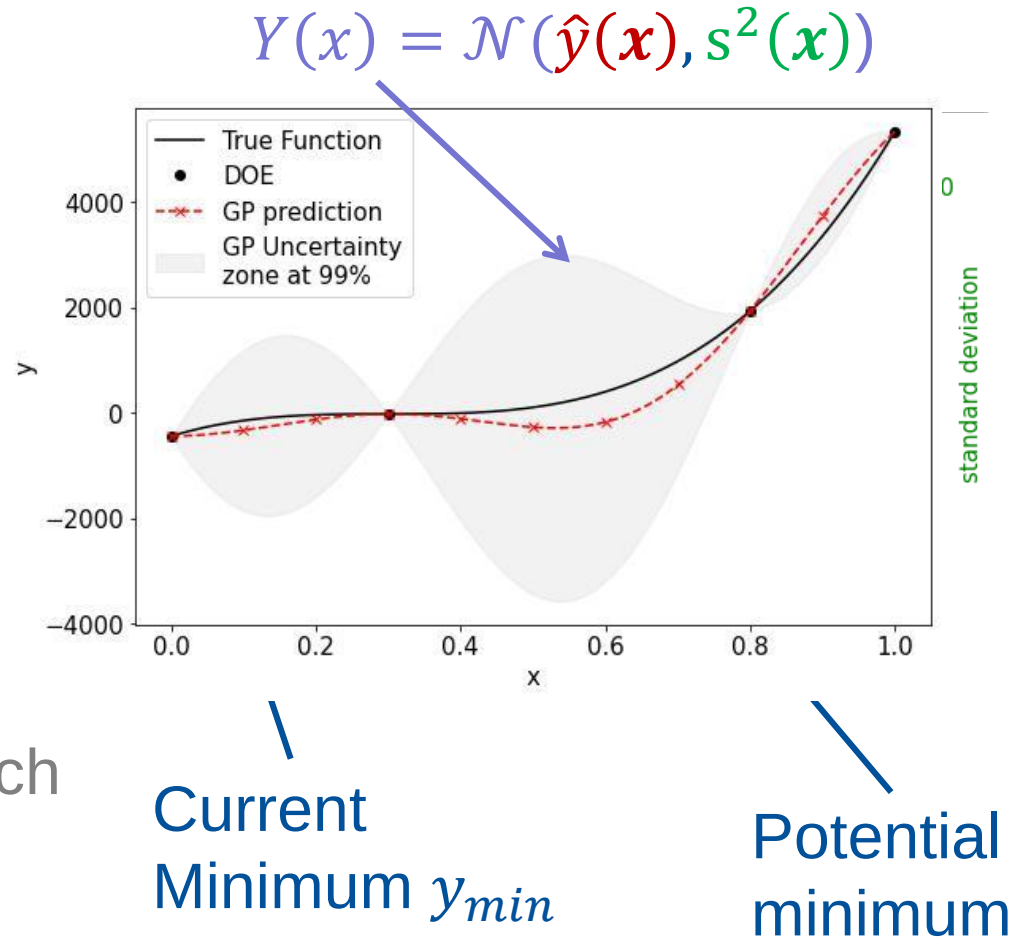
How to build an efficient iterative process?

- Find the global minimum with a limited budget of function evaluations
- Use Bayesian information to detect interesting and promising areas (exploitation/exploration trade-off)



Bayesian optimization

1. **Probabilistic model**
(surrogate model)
uses data and Bayes theorem to compute posterior distribution
2. **Optimization done via an acquisition function**
uses the posterior distribution to decide which data to obtain



Example: BO to tune NN hyperparameters within AlphaGo

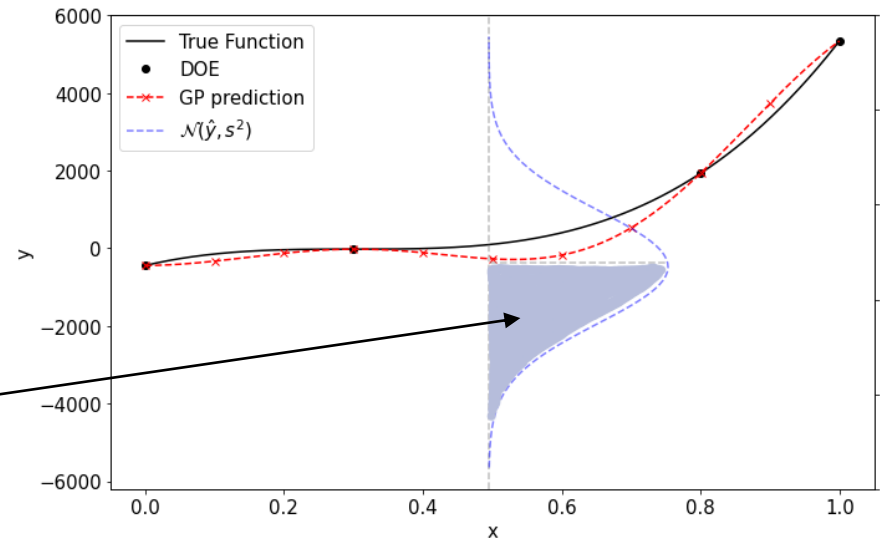
Chen, Y., Huang, A., Wang, Z., Antonoglou, I., Schrittwieser, J., Silver, D., & de Freitas, N. (2018). Bayesian optimization in alphago. *arXiv preprint arXiv:1812.06855*.

Enrichment infill sampling criterion

$$f(\mathbf{x}) \Rightarrow Y(\mathbf{x}) = \mathcal{N}(\hat{y}(\mathbf{x}), s^2(\mathbf{x}))$$

Measure of progress defined by the Improvement

$$I(\mathbf{x}) = \max(0, \underbrace{y_{\min} - Y(\mathbf{x})}_{>0 \text{ if } Y(\mathbf{x}) < y_{\min}})$$



$y_{\min}$ = current min of the database

Expected Improvement

$$EI(\mathbf{x}) = \mathbb{E}[I(\mathbf{x})] = \mathbb{E}[\max\{0, y_{\min} - Y(\mathbf{x})\}]$$

EGO for Efficient Global Optimization

Jones, D. R., Schonlau, M., and Welch, W. J., "Efficient global optimization of expensive black-box functions," Journal of Global optimization, Vol. 13, No. 4, 1998, pp. 455–492.

Enrichment infill sampling criterion

$$f(\mathbf{x}) \Rightarrow Y(\mathbf{x}) = \mathcal{N}(\hat{y}(\mathbf{x}), s^2(\mathbf{x})) \quad \text{Expected Improvement criterion (EI)}$$

Kriging or Gaussian process of the objective function

$$\text{EI}(\mathbf{x}) = \mathbb{E}[\max(y_{\min} - Y(\mathbf{x}), 0)]$$

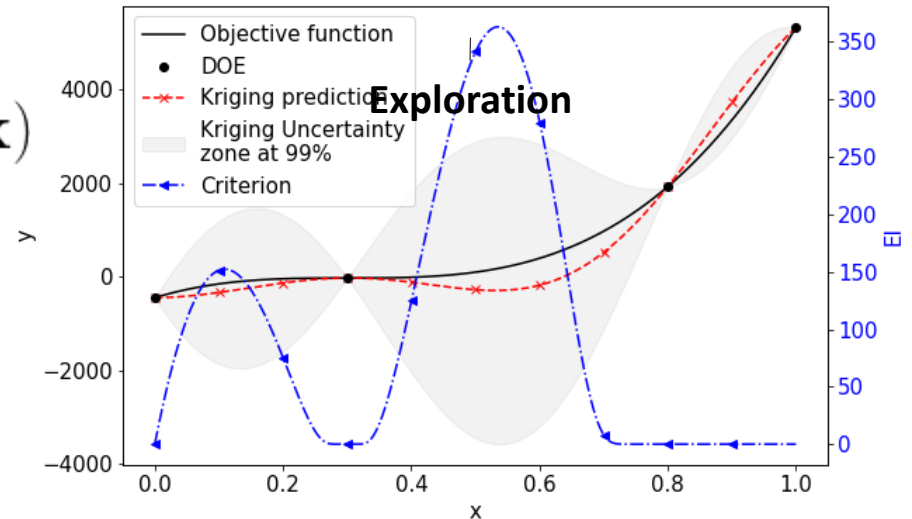
Φ cumulative distribution function
 ϕ probability density function of $\mathcal{N}(0,1)$

$$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \\ \text{s.t.} \\ \text{EI}(\mathbf{x}) = (y_{\min} - \hat{y}(\mathbf{x}))\Phi\left(\frac{y_{\min} - \hat{y}(\mathbf{x})}{s(\mathbf{x})}\right) + s(\mathbf{x})\phi\left(\frac{y_{\min} - \hat{y}(\mathbf{x})}{s(\mathbf{x})}\right) \\ c_1(\mathbf{x}) \leq 0 \\ \vdots \\ c_m(\mathbf{x}) \leq 0 \end{cases}$$

Surrogate models
 (objective & constraints)



$$\begin{cases} \text{Exploitation} \\ \max_{\mathbf{x} \in \mathbb{R}^d} \text{EI}(\mathbf{x}) \\ \text{s.t.} \\ \hat{c}_1(\mathbf{x}) \leq 0 \\ \vdots \\ \hat{c}_m(\mathbf{x}) \leq 0 \end{cases}$$

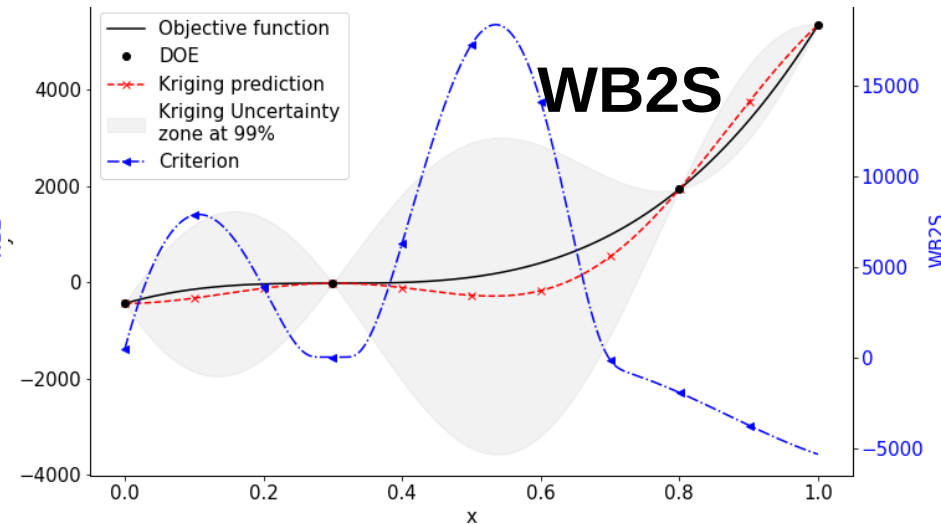
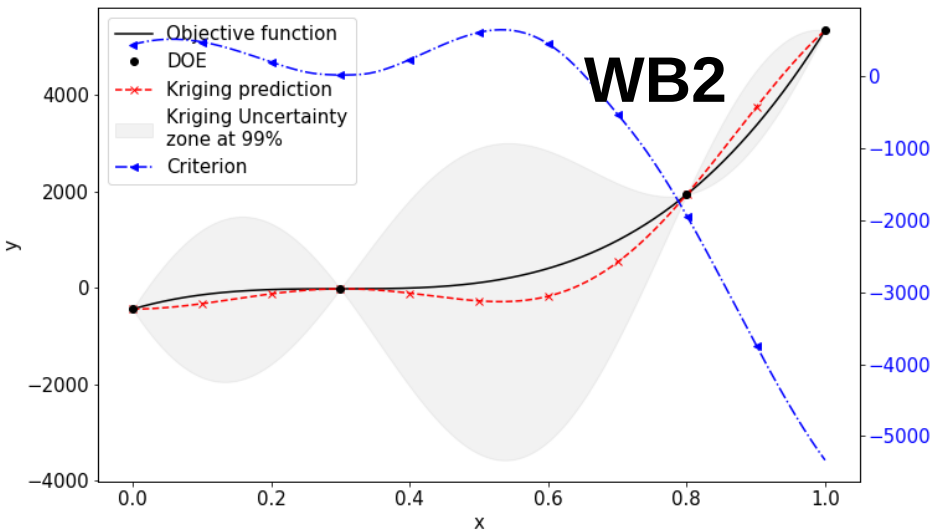
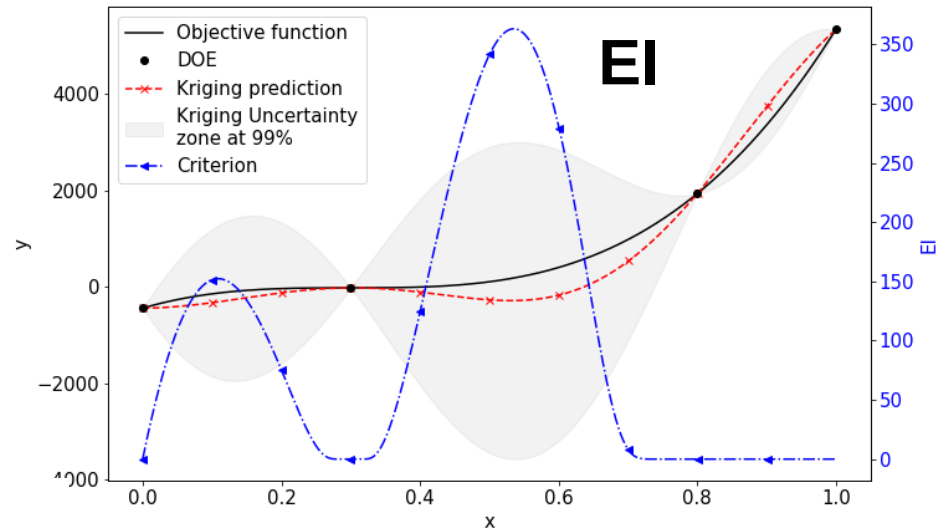
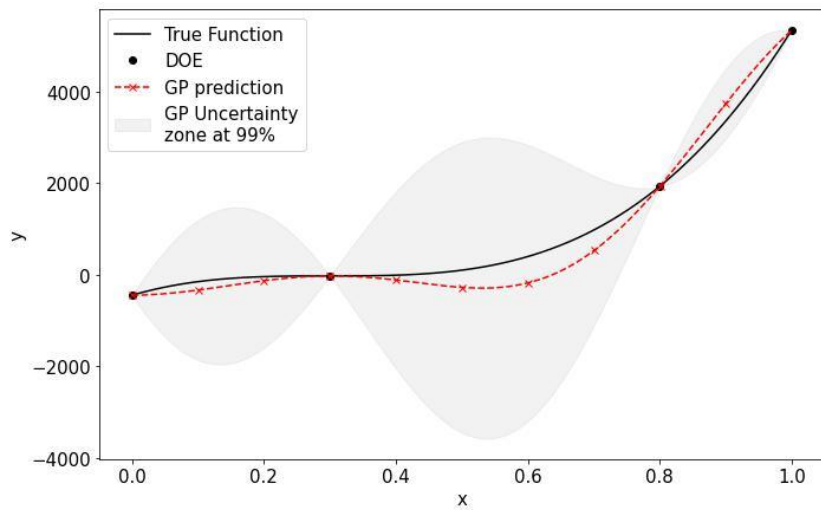


➔ Different criteria available for the acquisition function (EI, WB2, WB2S)

Jones, D. R., Schonlau, M., and Welch, W. J., "Efficient global optimization of expensive black-box functions," Journal of Global optimization, Vol. 13, No. 4, 1998, pp. 455–492.

Bartoli, N., Lefebvre, T., Dubreuil, S., Olivanti, R., Priem, R., Bons, N., Martins, J. R. A. A., Morlier, J. Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. Aerospace Science and Technology, 90:85–102, 2019.

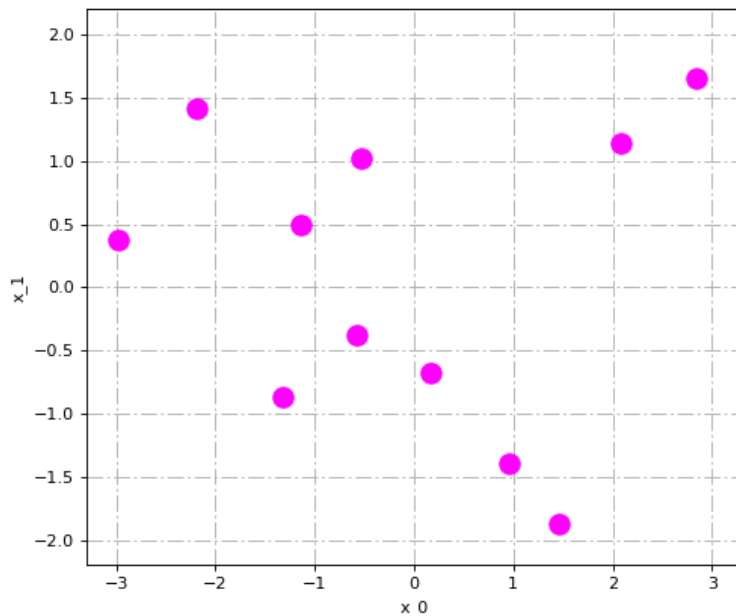
Mono objective: different criteria available



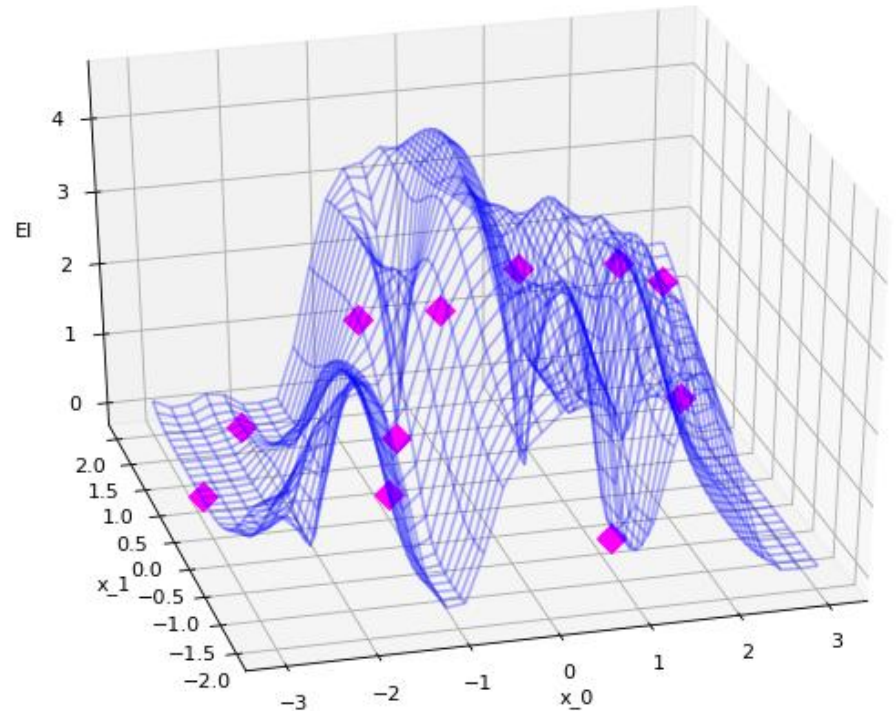
Enrichment for global optimization

$$\begin{cases} \max_{x \in \mathbb{R}^d} \alpha_f(x) & \text{max Acquisition function} \\ \text{s.t.} \\ \hat{c}_1(x) \leq 0 \\ \vdots \\ \hat{c}_m(x) \leq 0 \end{cases}$$

Local optimizer (Cobyla, Slsqp, Snopt) with ‘multi-start’ approach



● Initial DOE (LHS)

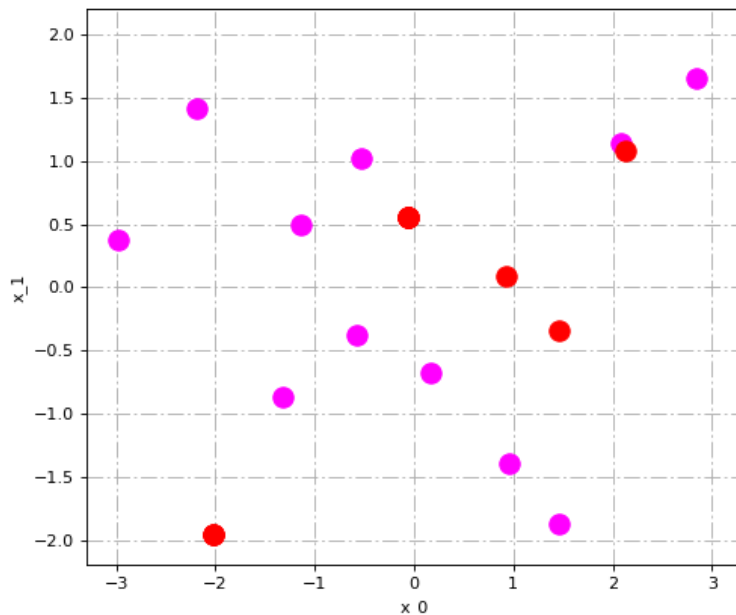


SNOPT: Sparse Nonlinear OPTimizer, Philip Gill, Walter Murray and Michael Saunders, Stanford Business Software Inc.

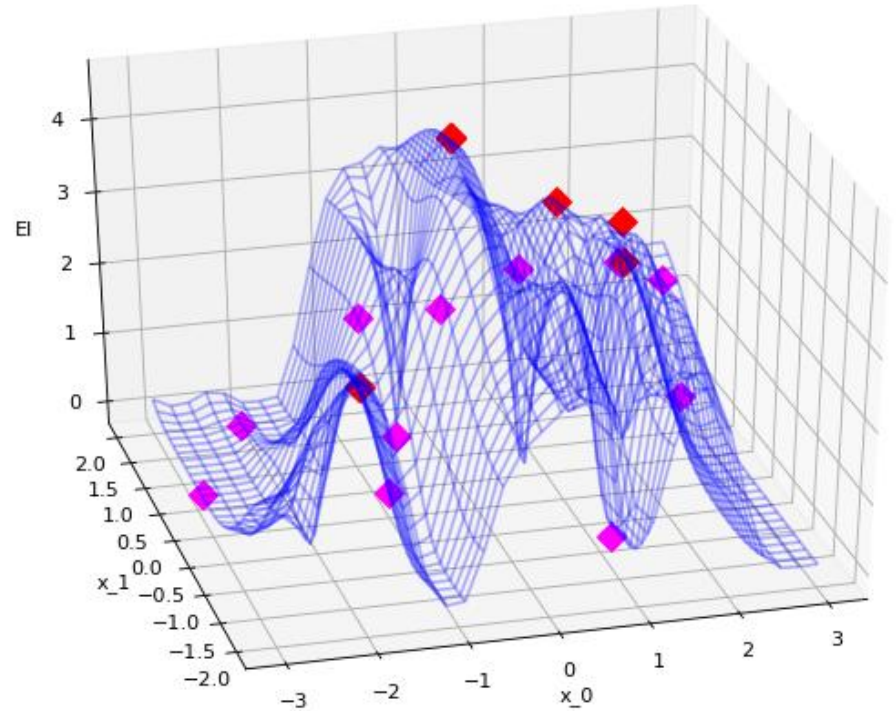
Enrichment for global optimization

$$\begin{cases} \max_{x \in \mathbb{R}^d} & \alpha_f(x) \\ \text{s.t.} & \\ & \hat{c}_1(x) \leq 0 \\ & \vdots \\ & \hat{c}_m(x) \leq 0 \end{cases}$$

Local optimizer (Cobyla, Slsqp, Snopt) with ‘multi-start’ approach



- Initial DOE(LHS)
- Optimization results

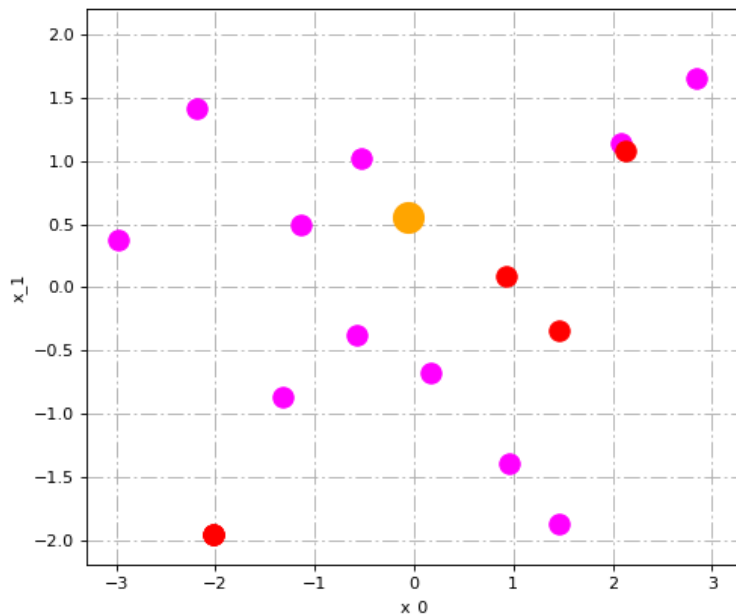


Sensitivity to starting points
Drives the global behaviour

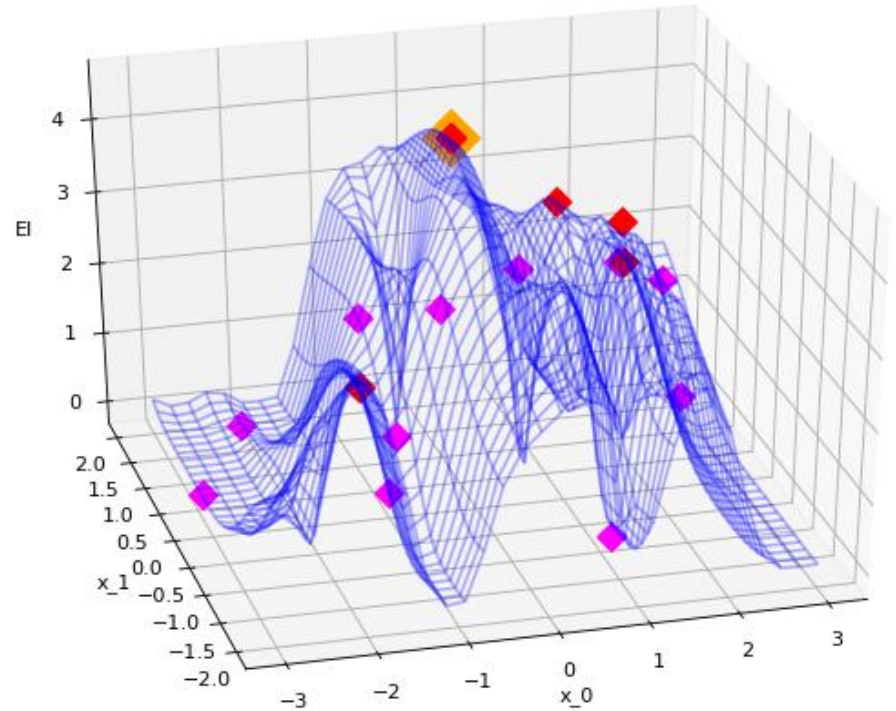
Enrichment for global optimization

$$\begin{cases} \max_{x \in \mathbb{R}^d} & \alpha_f(x) \\ \text{s.t.} & \\ & \hat{c}_1(x) \leq 0 \\ & \vdots \\ & \hat{c}_m(x) \leq 0 \end{cases}$$

Local optimizer (Cobyla, Slsqp, Snopt) with ‘multi-start’ approach



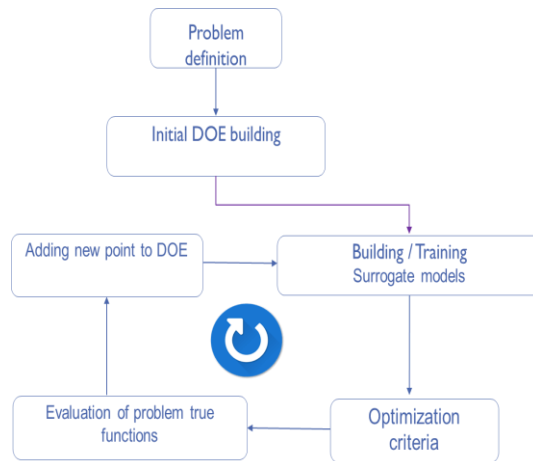
- Initial DOE(LHS)
- Optimization results
- Retained point



Sensitivity to starting points
Drives the global behaviour

SEGO

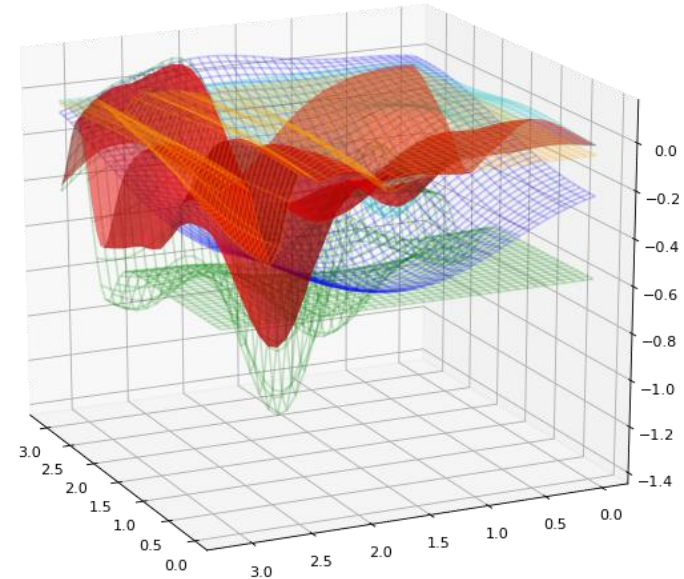
Super Efficient Global Optimization



Global optimization with
limited number of function
evaluations

MOE

Mixture Of Experts



Combination of GP
surrogate models

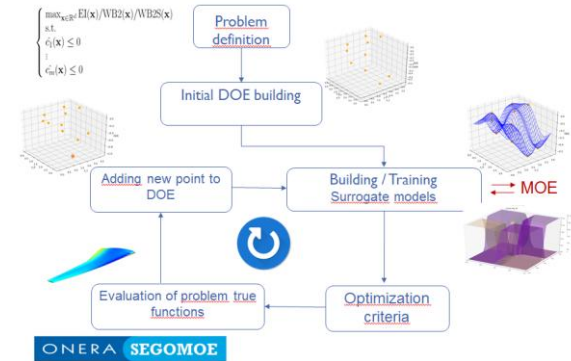


- Jordan, M. I., Jacobs, R. A., "Hierarchical mixtures of experts and the EM algorithm", Neural Comput. 6 (1994) 181–214.
- Bettebghor, D., Bartoli, N., Grihon, S., Morlier, J., and Samuelides, M., "Surrogate modeling approximation using a mixture of experts based on EM joint estimation," Structural and Multidisciplinary Optimization, Vol. 43, No. 2, 2011, pp. 243–259
- Liem, R. P., Mader, C. A., and Martins, J. R. R. A., "Surrogate Models and Mixtures of Experts in Aerodynamic Performance Prediction for Mission Analysis," Aerospace Science and Technology, Vol. 43, 2015, pp. 126–151

SEGOMOE main characteristics

$$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^d} & f(\mathbf{x}) \\ \text{s.t.} & \\ & c_1(\mathbf{x}) \leq 0 \cdots c_j(\mathbf{x}) = 0 \cdots c_m(\mathbf{x}) \leq 0 \end{cases}$$

d design variables
 m mixed constraints

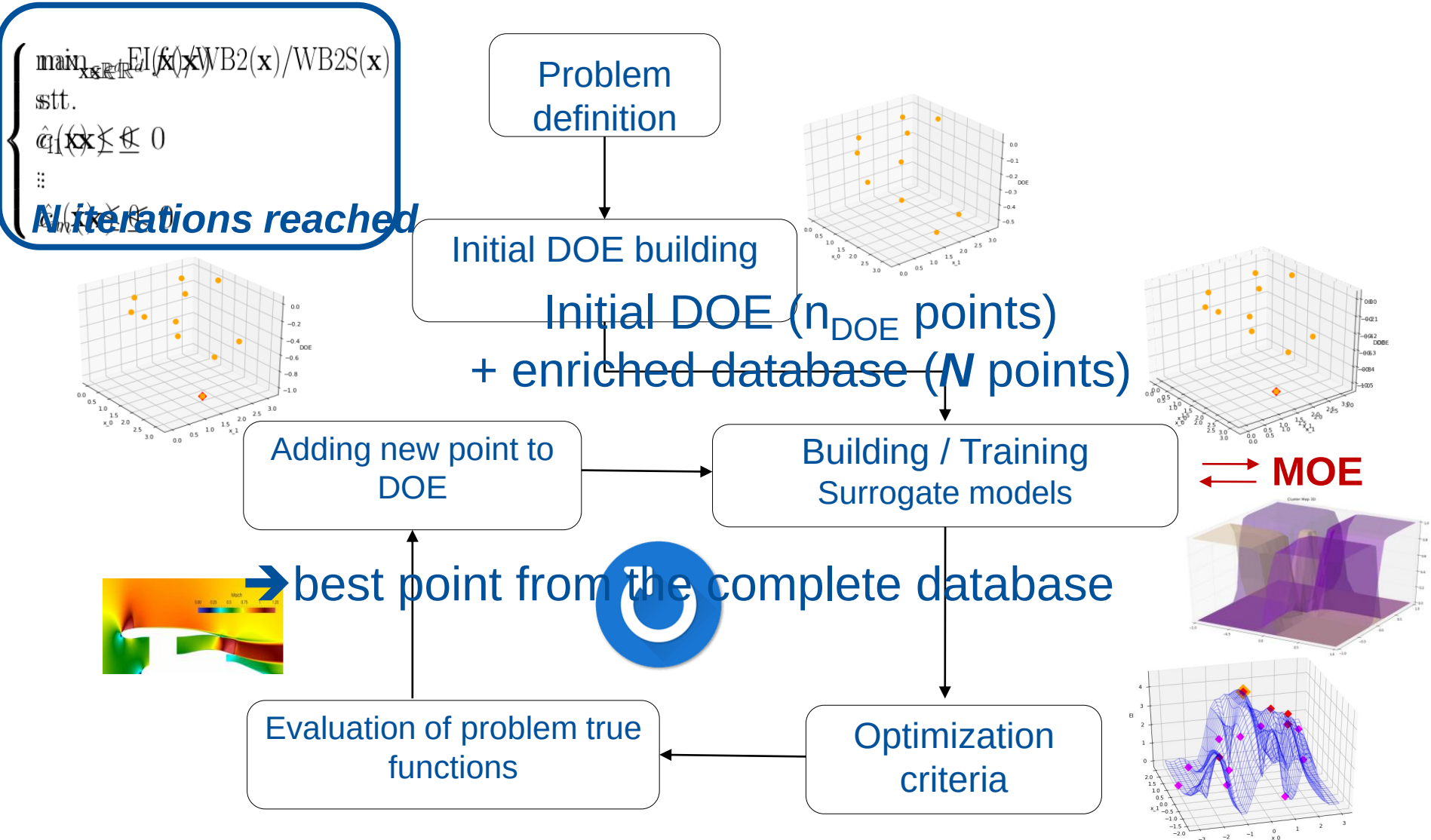


- Mono & **multi objective** Bayesian optimizer
- **Mono** & Multi fidelity sources
- Equality & inequality constraints (1 ~ 100 constraints)
- Intermediate dimension problem (1 ~ 100 variables)
- **Heterogenous variables** (continuous, discrete, categorical)
- Costly evaluation (CFD, FEM, objective and/or constraints)
- Handling non linear objectives & constraints (black box, no derivative available)
- Based on SMT toolbox for surrogate models
- Remote access via a web interface



Bartoli, N., Lefebvre, T., Dubreuil, S., Olivanti, R., Priem, R., Bons, N., Martins, J.R.R.A., Bouhlel, M.-A. Bouhlel & Morlier, J. (2019). Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. Aerospace Science and technology, 90, 85-102.

SEGOMOE algorithm – Mono objective

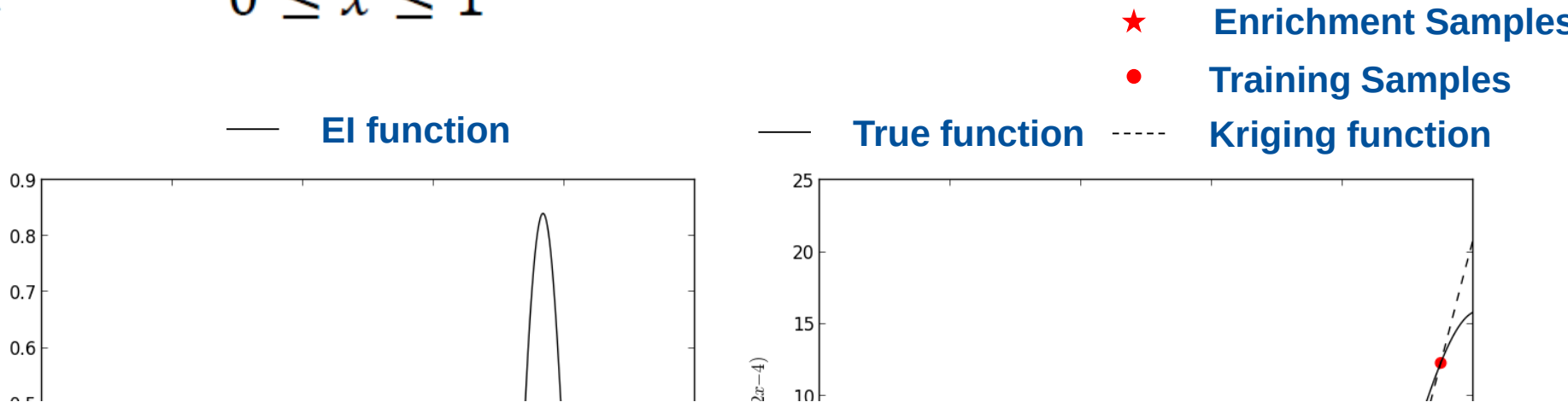


Bartoli, N., Lefebvre, T., Dubreuil, S., Olivanti, R., Priem, R., Bons, N., Martins, J.R.R.A., Bouhlef, M.-A. Bouhlef & Morlier, J. (2019). Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. *Aerospace Science and technology*, 90, 85-102.

Bayesian Optimization: Illustration on 1D example

$$\begin{cases} \min (6x - 2)^2 \sin(12x - 4) \\ s.t. \\ 0 \leq x \leq 1 \end{cases}$$

4 points for the initial DOE
6 iterations

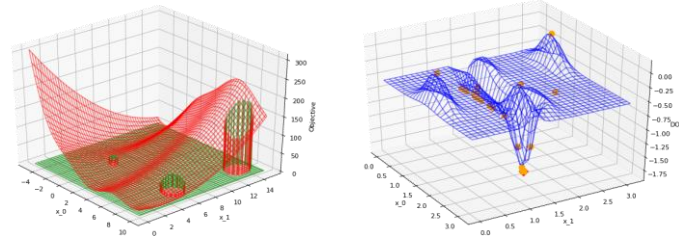


BO approach in the notebook

SEGOMOE validation – Mono Objective applications

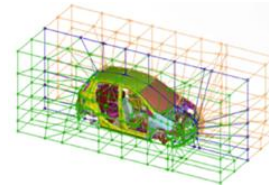
- Reference analytical functions

- Unconstrained multi-modal problems ($d=2, 5, 10$)
- Constrained problems ($d=10, m=8$)



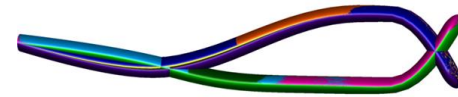
- MOPTA test case – automotive industry

$d = 50, m = 68$ Comparison with COBRA and COBYLA



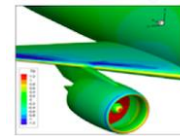
- Wing aerodynamic optimization (ADODG6)

$d = 17, m = 1$ Global optimum found compared to SNOPT



- Nacelle optimization

$d = 18, m = 2$ **AGILE H2020 project**
Better convergence than DAKOTA



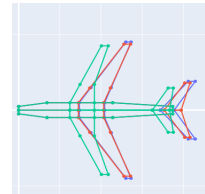
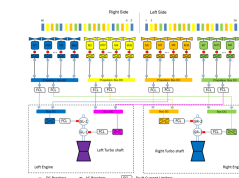
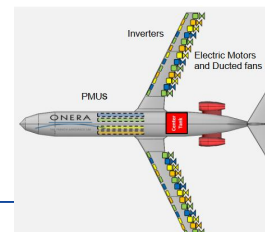
- Bombardier Research Aircraft Configuration use case

$d = 12, m = 8$
3h with SEGOMOE
> 8h with some internal optimizers (Isight)



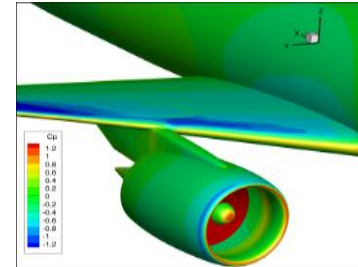
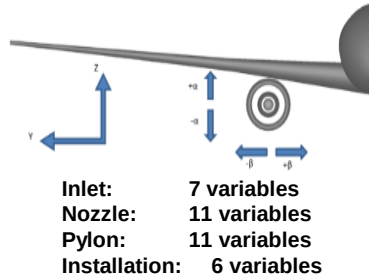
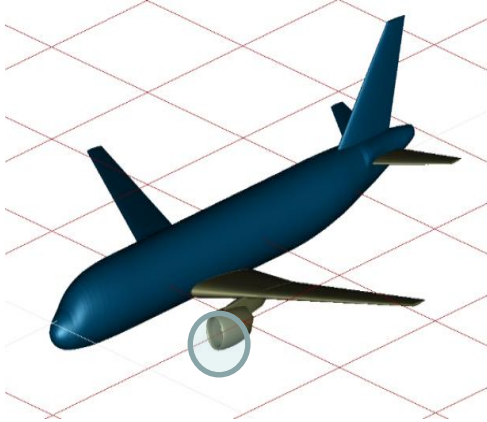
- Dragon hybrid electric aircraft

$d = 29$ (categorical choices) $m=5$

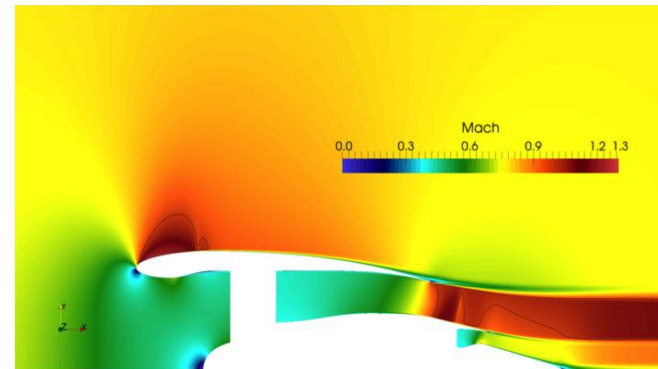
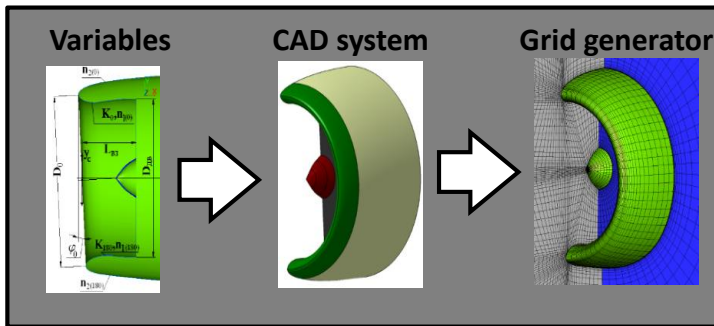


Nacelle Design for Large Regional jet

EU H2020 project, 2015-2018



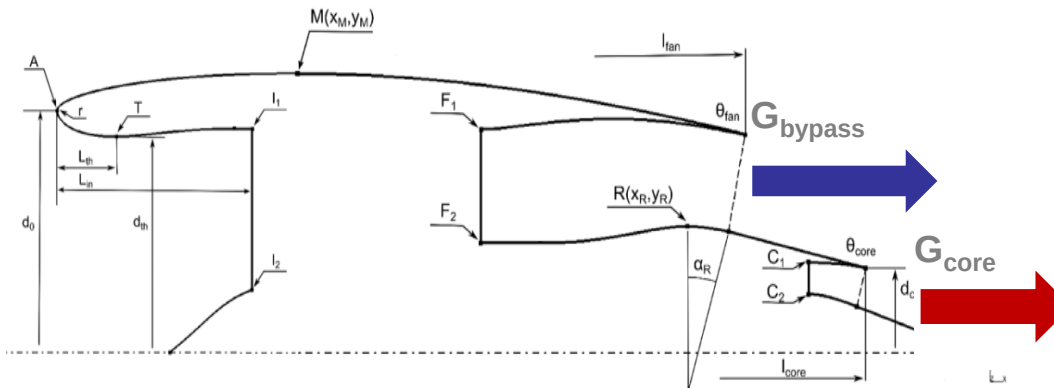
AIRCRAFT 3RD GENERATION MDO
 FOR INNOVATIVE COLLABORATION
 OF HETEROGENEOUS TEAMS OF EXPERTS



CFD solver for RANS equations on structured grid
 The solver is based on the finite-volume numerical method

Nacelle Design for Large Regional jet

- **Optimization problem:**
- 17 design variables (7 for the inlet, 10 for the nozzle)
- 1 objective function dP_{eff} (thrust-minus-drag losses),
- 2 inequality constraints for the mass flow rates



- **CFD computations** (EWT code from Tsagi)
 - Axisymmetric grid, $N = 59136$ cells
 - Regime: $M = 0.78$, $H = 11$ km (Cruise)
 - BPR = 9 and BPR = 12
 - 1 run: 30 minutes with 48 cores

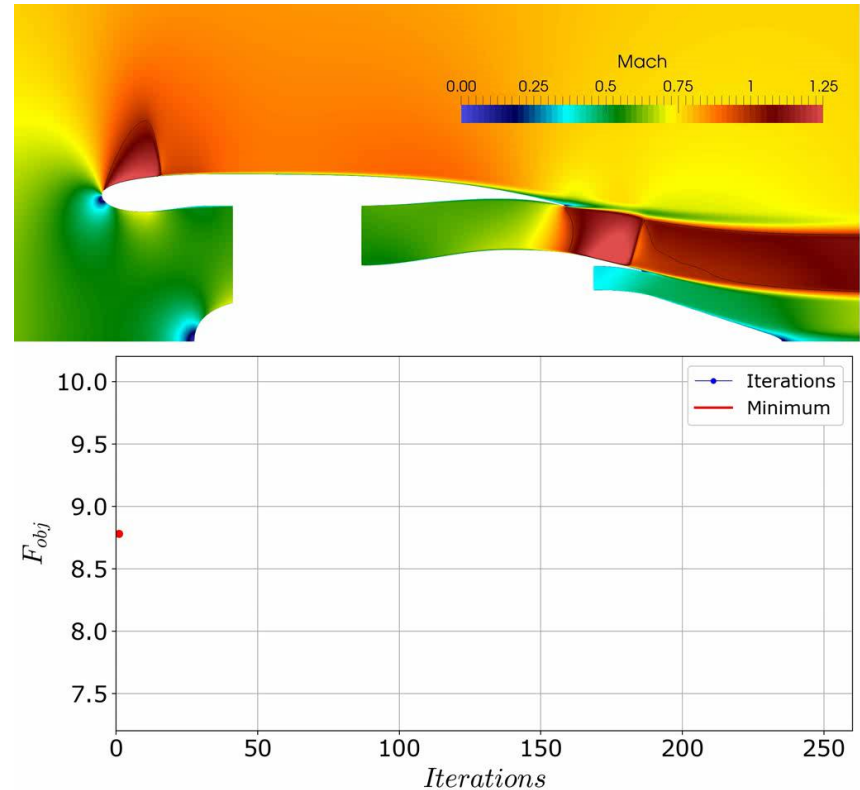
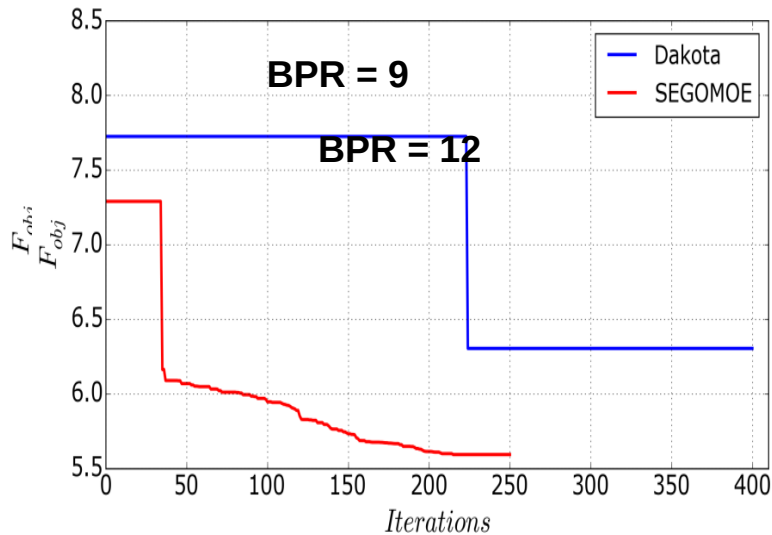


$$\left\{ \begin{array}{l} \min_{x \in \mathbb{R}^{17}} dP_{\text{eff}}(x) \\ \text{s.t.} \\ \left| 1 - \frac{G_{\text{core}}}{G_{\text{target}}^{\text{core}}} \right| < \epsilon_{\text{core}} \\ \left| 1 - \frac{G_{\text{bypass}}}{G_{\text{target}}^{\text{bypass}}} \right| < \epsilon_{\text{bypass}} \end{array} \right.$$

Nacelle design optimization with SEGOMOE

- Initial DOE of 30 points
- BPR = 12
- Mach number distribution

➔ Similar process for different BPR values



N. Bartoli, T. Lefebvre, S. Dubreuil, M. Panzeri, R. d'Ippolito, K. Anisimov, A. Savelyev, *Robust Nacelle Optimization design investigated in the AGILE European Project*, 19th AIAA/ISSMO Multidisciplinary Analysis and Optimization Conference, AIAA AVIATION Forum, (AIAA 2018-3250)

Recent methodological developments

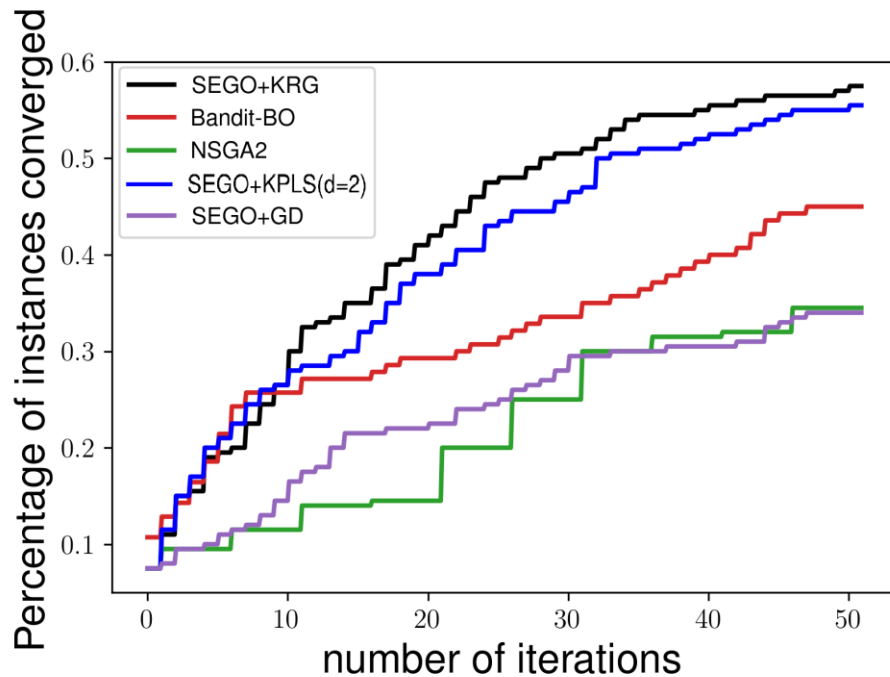
SEGOMOE capabilities

- To handle a large number of design variables
 ➔ KPLS based models
- To handle heterogeneous functions
 ➔ Mixture of experts models
- To handle highly non-convex constraints
 ➔ Adapted acquisition function
- To handle mixed integer variables
 ➔ Continuous relaxation & KPLS models
- To handle multifidelity models
 ➔ 2-step approach based on multifidelity Kriging
- To handle multiple objectives
 ➔ predicted Pareto Front approach

Validation for mixed integer variables

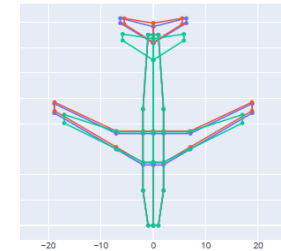
- 10 analytical tests cases (20 instances each)
⇒ 2 to 14 dimensions, with and without constraint

<https://mixed-optimization-benchmark.github.io/cases/>



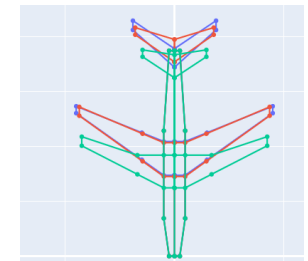
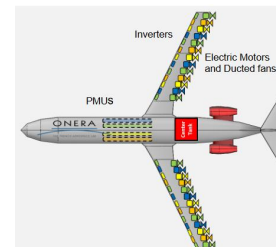
- 2 aircraft design problems with mixed continuous-categorical variables:

CeRAS: 14 dimensions



DRAGON: 29 dimensions

Distributed propulsion

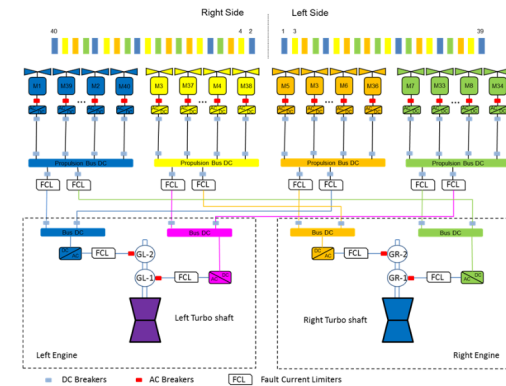
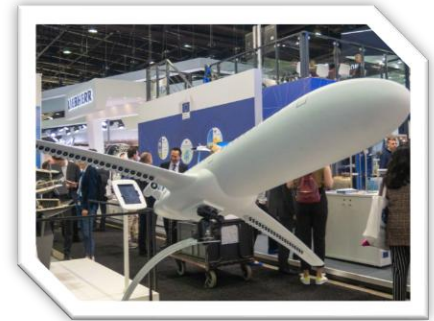
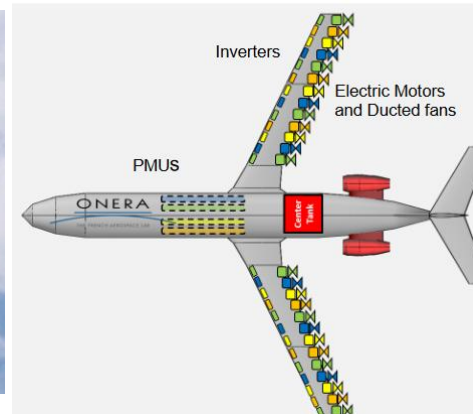


Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In *AIAA SCITECH 2022 Forum* (p. 0082).

Application to mixed categorical

Optimization problem DRAGON green aircraft concept

- ✓ 30% reduction of CO2 emissions by 2035
- ✓ Distributed electric propulsion aircraft: propulsive efficiency
- ✓ 150 passengers over 2750nm
- ✓ Transonic cruise speed (M0.78)

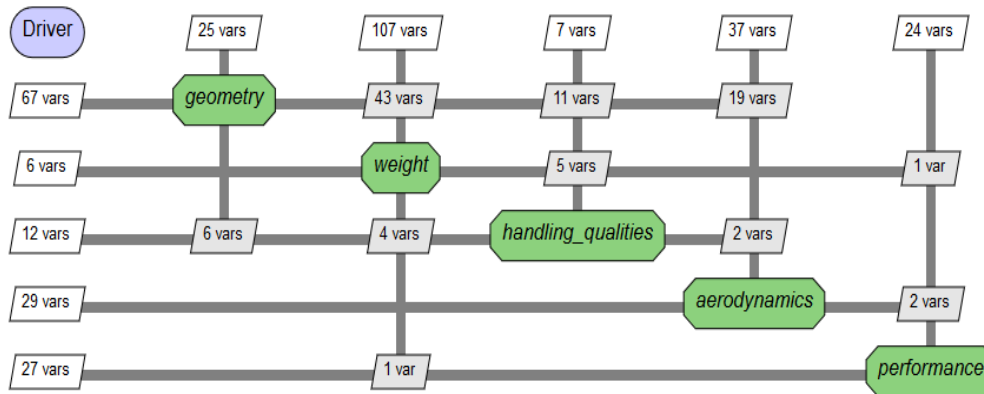


Phd P. Saves in collaboration with E. Nguyen Van, C. David, S. Defoort

P. Schmollgruber, C. Doll, J. Hermetz, R. Liaboeuf, M. Ridet, I. Cafarelli, O. Atinault, C. Francois, and B. Paluch. "Multidisciplinary Exploration of DRAGON: an ONERA Hybrid Electric Distributed Propulsion Concept". In: AIAA Scitech 2019, 2019

FAST-OAD: an OpenMDAO based aircraft sizing tool

Code overview



<https://github.com/fast-aircraft-design/FAST-OAD>

FAST OAD

Future Aircraft Sizing Tool - Overall Aircraft Design

FastOAD* conceptual design framework:

- OpenSource framework developed by ONERA/ISAE-SUPAERO
- Based on [OpenMDAO](#) 
- Automates MDA/MDO for simple and rapid OAD studies, concept evaluation and optimization
- Includes [Level 0 disciplinary models](#) for transport aircraft (geometry, weight, HQ, aerodynamics, mission/performance...)
- Modularity of each discipline model to include higher fidelity modelling

Christophe David, Scott Delbecq, Sébastien Defoort, Peter Schmollgruber, Emmanuel Benard, Valérie Pommier-Budinger, From FAST to FAST-OAD: An open source framework for rapid Overall Aircraft Design, 2021 IOP Conf. Ser.: Mater. Sci. Eng.1024 012062.

DRAGON green aircraft concept

	Function/variable	Nature	Quantity	Range
Minimize	Fuel mass	cont	1	
with respect to	Fan operating pressure ratio	cont	1	[1.05, 1.3]
	Wing aspect ratio	cont	1	[8, 12]
	Angle for swept wing	cont	1	[15, 40] (°)
	Wing taper ratio	cont	1	[0.2, 0.5]
	HT aspect ratio	cont	1	[3, 6]
	Angle for swept HT	cont	1	[20, 40] (°)
	HT taper ratio	cont	1	[0.3, 0.5]
	TOFL for sizing	cont	1	[1800., 2500.] (m)
	Top of climb vertical speed for sizing	cont	1	[300., 800.] (ft/min)
	Start of climb slope angle	cont	1	[0.075., 0.15.] (rad)
	Total continuous variables		10	
	Architecture	cat	17 levels	{1,2,3, ..., 15,16,17}
	Turboshaft layout	cat	2 levels	{1,2}
	Total categorical variables		2	
	Total relaxed variables		29	
subject to	Wing span < 36 (m)	cont	1	
	TOFL < 2200 (m)	cont	1	
	Wing trailing edge occupied by fans < 14,4 (m)	cont	1	
	Climb duration < 1740 (s)	cont	1	
	Top of climb slope > 0.0108 (rad)	cont	1	
	Total constraints		5	



- 10 continuous design variables
- 2 categorical design variables

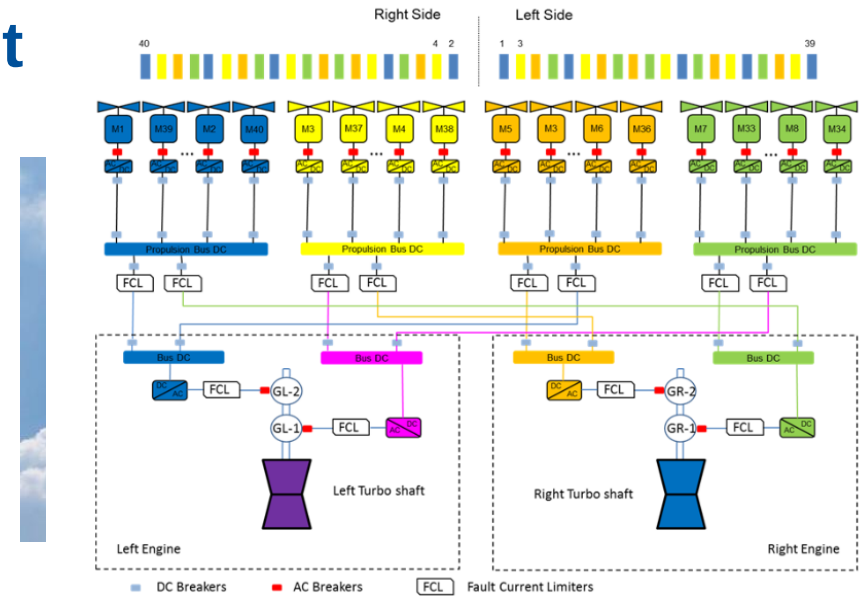
- 5 inequality constraints (MC)
- Fuel mass to minimize

Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In *AIAA SCITECH 2022 Forum* (p. 0082).

DRAGON green aircraft concept

Architecture	cat	17 levels	{1,2,3, ..., 15,16,17}
Turboshaft layout	cat	2 levels	{1,2}
Total categorical variables		2	
Total relaxed variables		29	

Architecture number	number of motors	number of generators
1	8	2
2	12	2
3	16	2
4	20	2
5	24	2
6	28	2
7	32	2
8	36	2
9	40	2
10	8	4
11	16	4
12	24	4
13	32	4
14	40	4
15	12	6
16	24	6
17	36	6



position	y ratio	tail	VT aspect ratio	VT taper ratio
under wing	0.25	without T-tail	1.8	0.3
behind	0.34	with T-tail	1.2	0.85

17x2 possibilities → 34 associated optimization problems to solve 😞
 or use dedicated kernels in your GP models → 1 optimization problem to solve 😊

Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In *AIAA SCITECH 2022 Forum* (p. 0082).

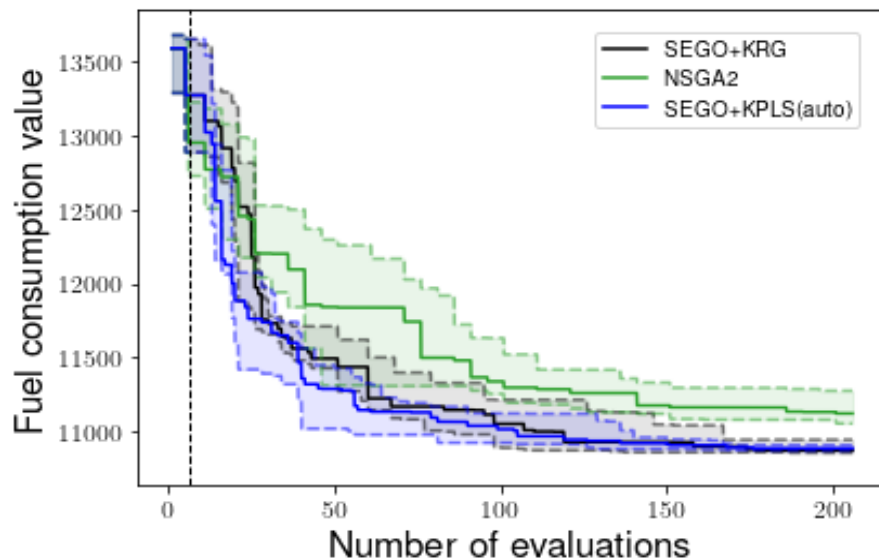
DRAGON optimization results

PhD P. Saves 2020-2023 (ISAE-ONERA)

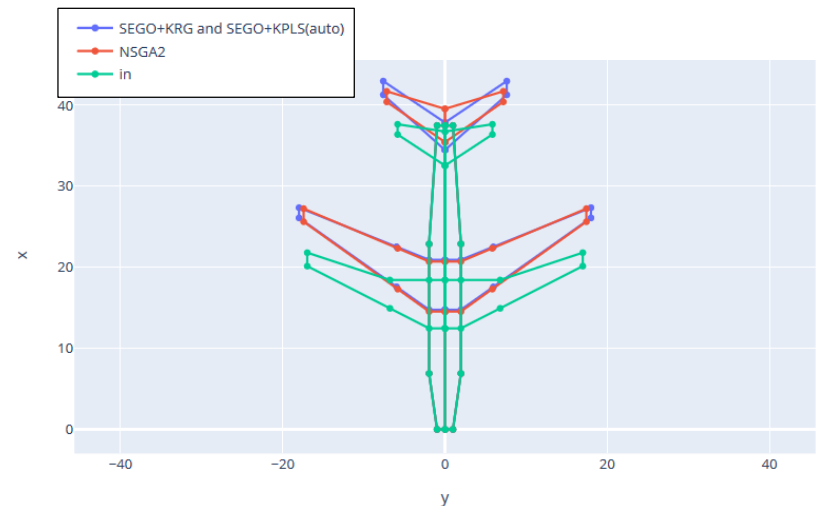


Convergence plots for DRAGON

10 runs



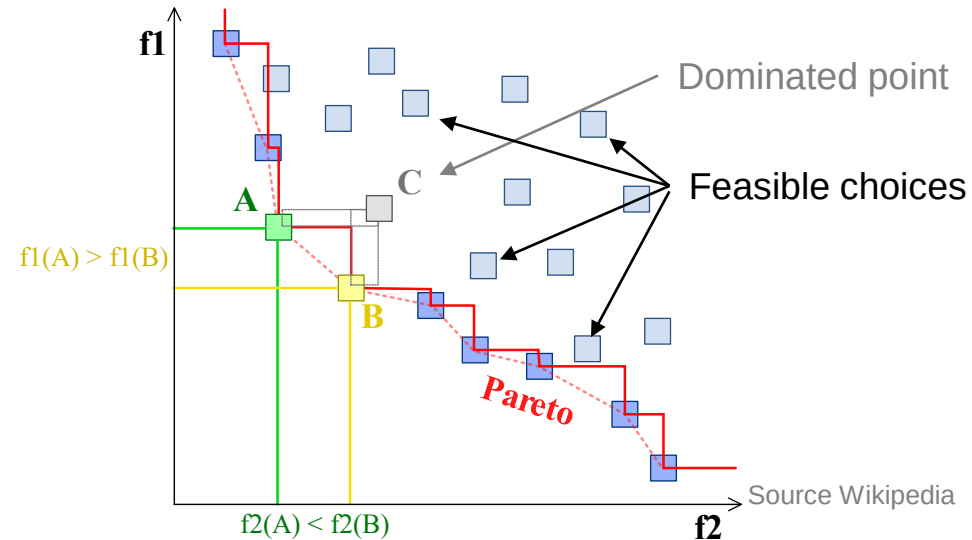
Optimal results



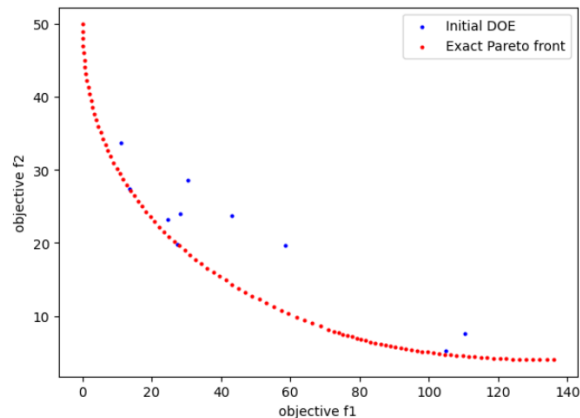
Saves, P., Bartoli, N., Diouane, Y., Lefebvre, T., Morlier, J., David, C., ... & Defoort, S. (2022). Bayesian optimization for mixed variables using an adaptive dimension reduction process: applications to aircraft design. In *AIAA SCITECH 2022 Forum* (p. 0082).

How to perform a multiobjective optimization?

$$\begin{cases} \min_{\mathbf{x} \in \mathbb{R}^d} & \mathbf{f} = [f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_n(\mathbf{x})] \\ \text{s.t.} & \\ c_1(\mathbf{x}) \leq 0 & \\ \vdots & \\ c_m(\mathbf{x}) \leq 0 & \end{cases} \quad \mathbf{n} \text{ objectives}$$



Build a Pareto Front (non dominated points)

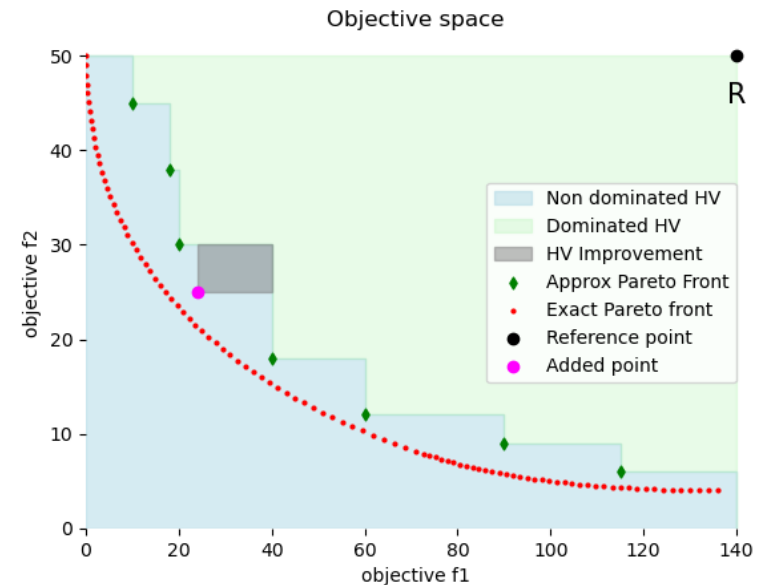
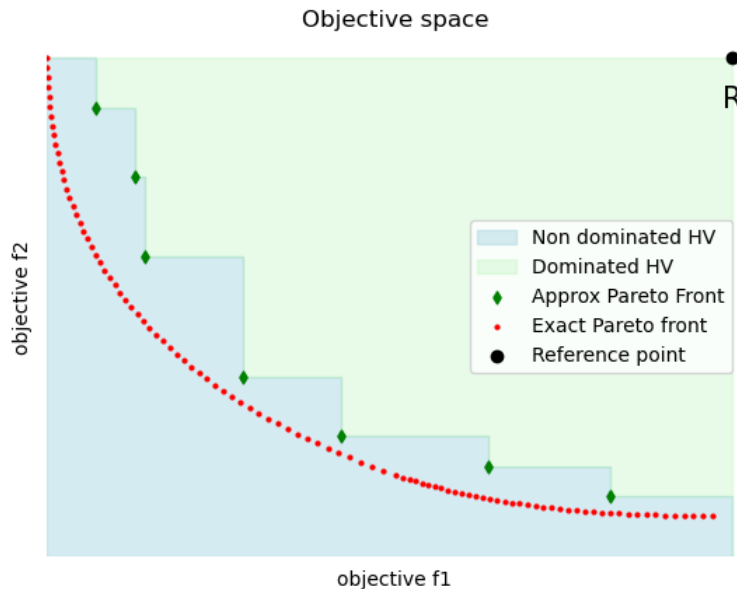
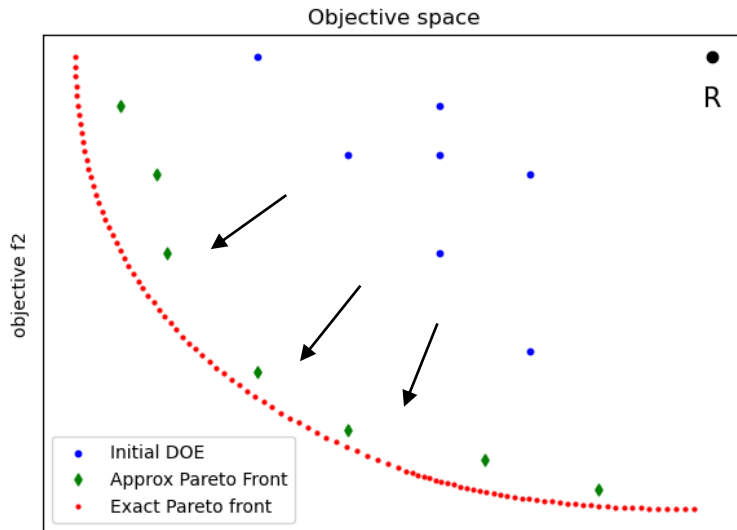


Internship R. Grapin 2021-2022

Multi objective: HyperVolume Improvement

From the initial DOE
add new points in order to
improve the Hyper Volume

Infill criterion based on Gaussian
Process each objective i : $\mathcal{N}(\hat{y}_i(x), s_i^2(x))$



New infill criterion proposed for Multi-objective

$$f(\mathbf{x}) \Rightarrow Y(\mathbf{x}) = \mathcal{N}(\hat{\mathbf{y}}(\mathbf{x}), s^2(\mathbf{x}))$$

- Mono-objective

$$\text{WB2S}(\mathbf{x}) = s \text{EI}(\mathbf{x}) - \hat{\mathbf{y}}(\mathbf{x})$$

scaling factor

- Multi-objective

$$\alpha_f^{\text{reg}}(\mathbf{x}) = \gamma \alpha_f(\mathbf{x}) - \psi(\mu_f(\mathbf{x}))$$

Acquisition function

$$\alpha_f(\mathbf{x}) = \text{EHVI}(\mathbf{x}), \text{PI}(\mathbf{x}), \text{MPI}(\mathbf{x}), \dots$$

based on Hyper Volume Improvement

Regularized choices

$$\left\{ \begin{array}{ll} (\text{reg} = \max) & : \quad \psi(\hat{\mathbf{y}}(\mathbf{x})) = \max_{i \leq n} \hat{y}_i(\mathbf{x}) \\ (\text{reg} = \text{sum}) & : \quad \psi(\hat{\mathbf{y}}(\mathbf{x})) = \sum_{i=1}^n \hat{y}_i(\mathbf{x}) \end{array} \right.$$

Bartoli, N., Lefebvre, T., Dubreuil, S., Olivanti, R., Priem, R., Bons, N., Martins, J. R. A. A., Morlier, J. Adaptive modeling strategy for constrained global optimization with application to aerodynamic wing design. Aerospace Science and Technology, 90:85–102, 2019.

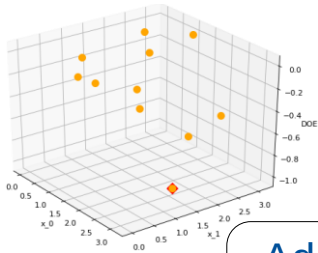
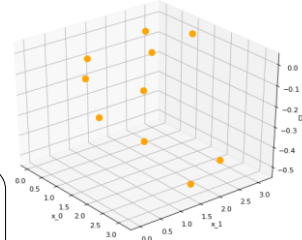
Grapin, R., Diouane, Y., Morlier, J., Bartoli, N., Lefebvre, T., Saves, P., & Bussemaker, J. H. (2022). Constrained Multi-Objective Bayesian Optimization with Application to Aircraft Design. In AIAA AVIATION 2022 Forum (p. 4053)

SEGOMOE algorithm – Multi objective

$$\begin{cases} \min_{x \in \mathbb{R}^d} & f = [f_1(x), f_2(x), \dots, f_n(x)] \\ \text{s.t.} & \\ & c_1(x) \leq 0 \quad \text{\textit{n objectives}} \\ & \vdots \\ & c_m(x) \leq 0 \end{cases}$$

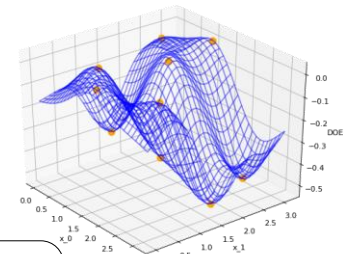
Problem
definition

Initial DOE building

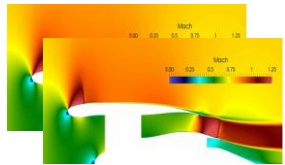
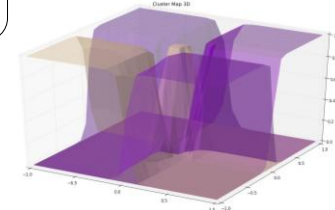


Adding new point to
DOE

Building / Training
Surrogate models

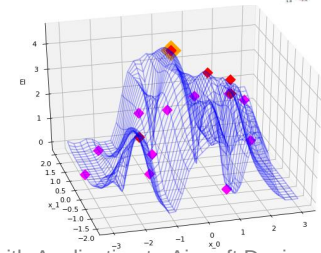


MOE SMT



Evaluation of problem true
functions

Optimization
criteria



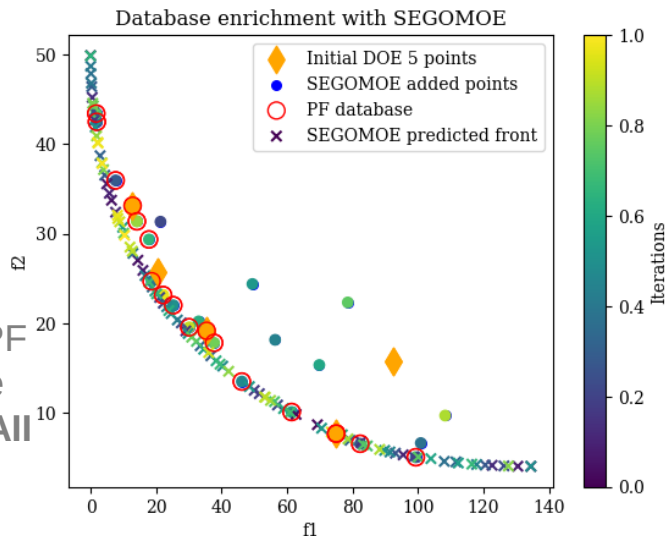
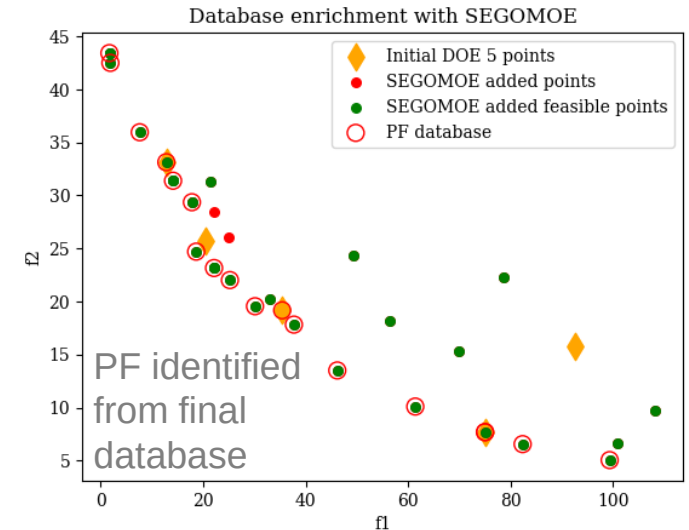
Grapin, R., Diouane, Y., Morlier, J., Bartoli, N., Lefebvre, T., Saves, P., & Bussemaker, J. H. (2022). Constrained Multi-Objective Bayesian Optimization with Application to Aircraft Design. In AIAA AVIATION 2022 Forum (p. 4053)

SEGOMOE algorithm – Multi objective

N iterations reached

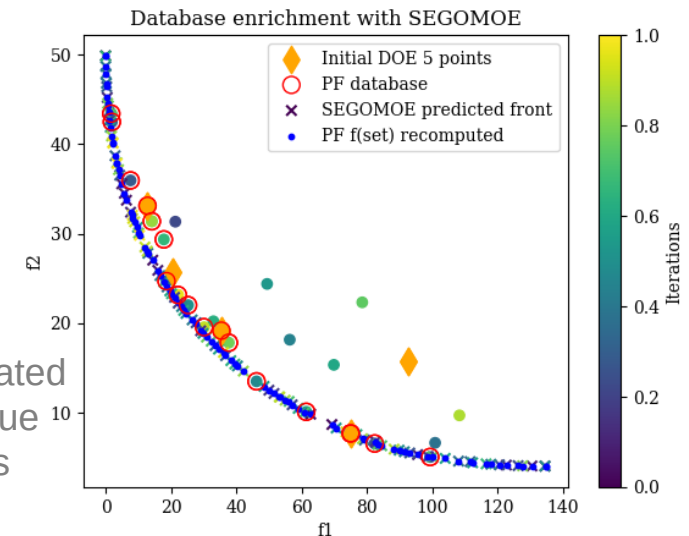
Initial DOE (n_{DOE} points)
+ enriched database (N points)

Build final Surrogate models
from these ($n_{\text{DOE}} + N$) points



→ Predicted PF
from surrogate
models (NSGAI
algo)

→ Re-evaluated
PF from true
functions



Outline

- Kriging based surrogate models
- Bayesian optimization
 - mono & multiobjective
- **AGILE 4.0 applications**

In collaboration with R. Lafage, J. H. Bussemaker, G. Donelli, J. M. Gomes de Mello, M. Mandorino, P. Della Vecchia



UNIVERSITÀ DEGLI STUDI DI NAPOLI
FEDERICO II

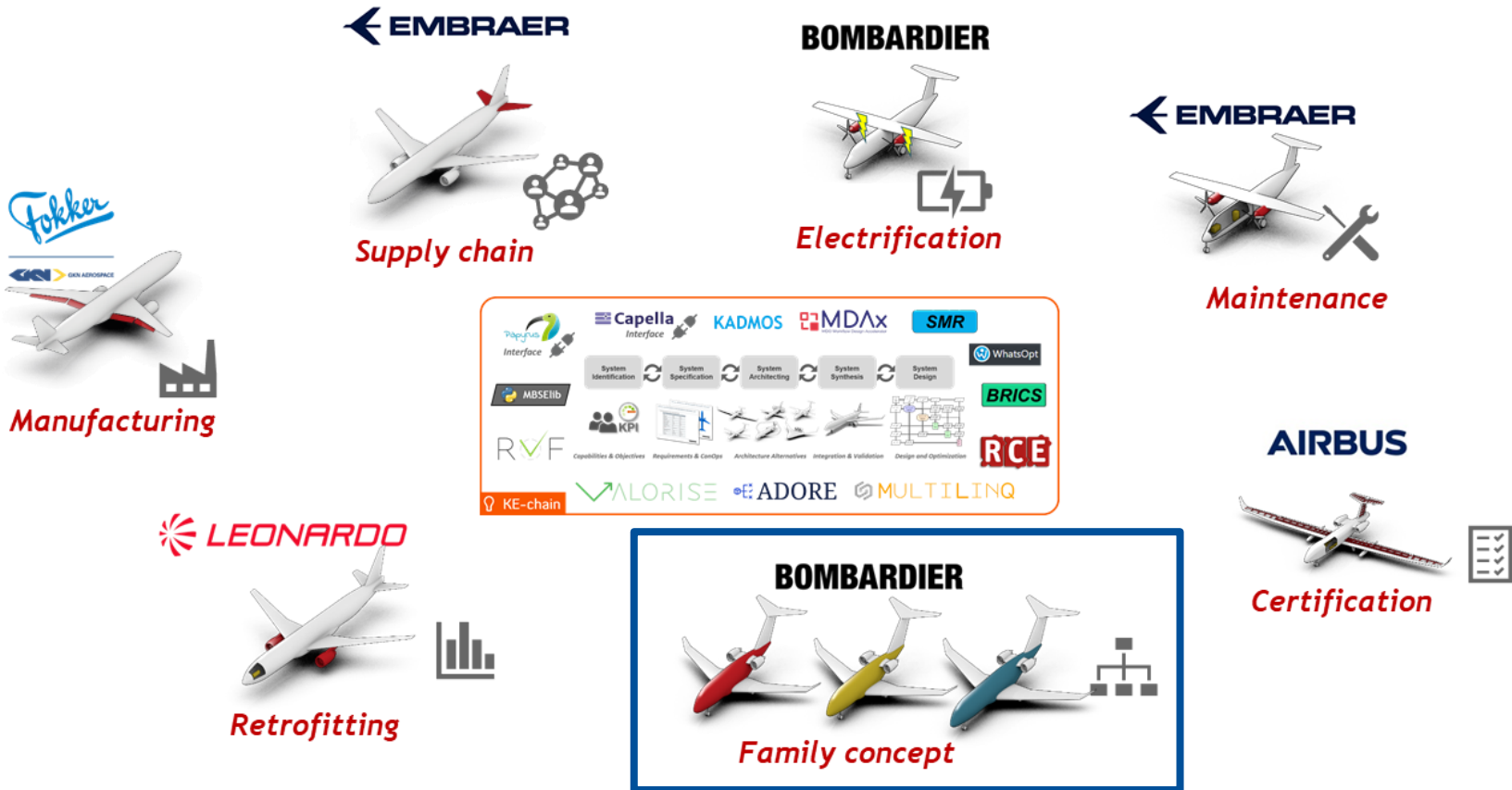
AGILE 4.0 

The AGILE 4.0 logo, with 'AGILE' in large blue letters and '4.0' in red. To the right are several small icons representing different aspects of Industry 4.0, including a gear, a factory, a person, and a network.

SEGOMOE validation – Multi Objective applications



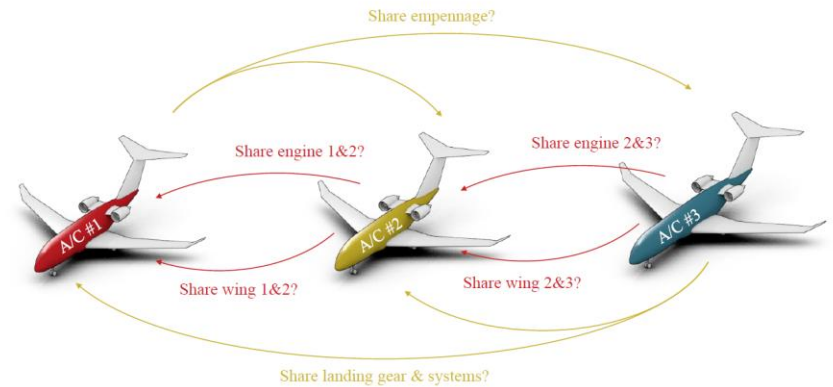
- AGILE4.0 application cases (2019-2023)



with inputs from
Jasper Bussemaker (DLR) and others AGILE 4.0 partners

Family concept design

- Tradeoff between commonality and performance
- Application: business jet family
 - Use case owner: Bombardier
 - Three 8-pax business jets
 - Variations in range and cabin length



- Share systems, landing gear, wings, engines, empennage
- One wing platform per-member basis (conforming to commonality selection)

BOMBARDIER

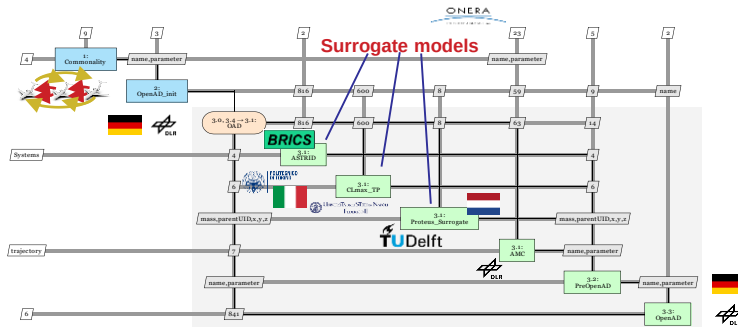
UNIVERSITÀ DEGLI STUDI DI NAPOLI FEDERICO II

Politecnico di Torino

DLR

TU Delft

ONERA
THE FRENCH AEROSPACE LAB



MDA: 1 hour for a family

No sharing: high performance



Max sharing: low investment costs

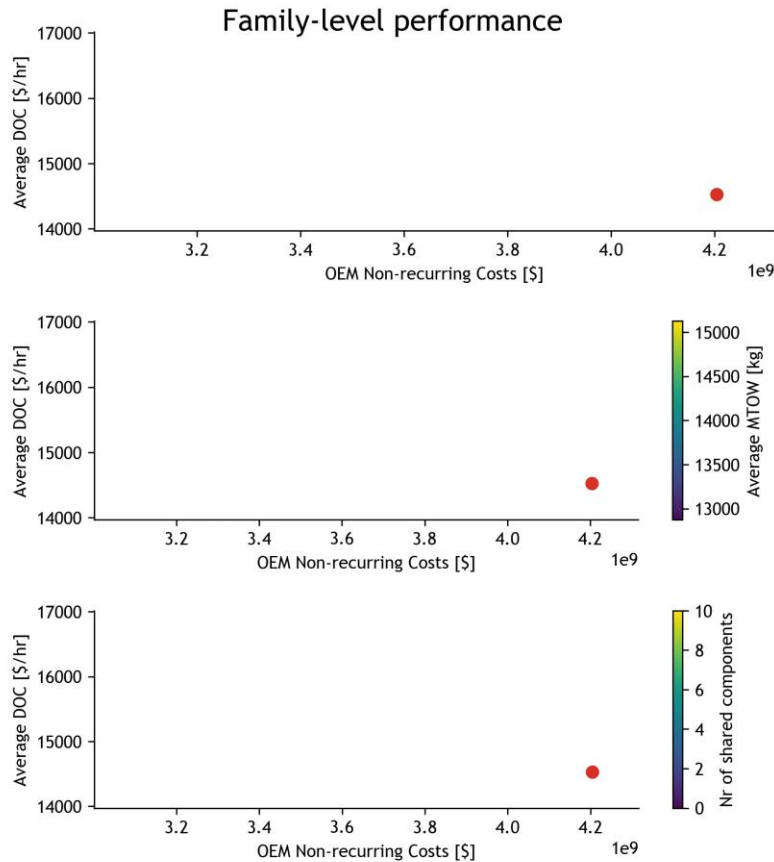


Select the best family with commonality choices (wing, engine, empennage)

Objectives	2 obj: Min (Direct Operating Costs, OEM Non – Recurring Costs)
Design variables	10 categorical: commonality with 2 levels 9 continuous: Leading edge sweep, rear spar location, Wing t/c for each family member
Constraints	2 ineq. : Balanced Field Length, Landing Field Length

Bussemaker, J. H., Ciampa, P. D., Singh, J., Fioriti, M., Cabaleiro De La Hoz, C., Wang, Z., ... & Mandorino, M. (2022). Collaborative Design of a Business Jet Family Using the AGILE 4.0 MBSE Environment. In AIAA Aviation 2022 Forum (p. 3934).

Family concept design



BOMBARDIER



Deutsches Zentrum
für Luft- und Raumfahrt



Business Jet Family #1 (DOE)

Design variables

	AC 1	AC 2	AC 3
Wing sized for	AC 1	AC 2	AC 3
Engine sized for	AC 1	AC 2	AC 3
Ldg gear sized for	AC 1	AC 2	AC 3
Systems sized for	AC 1	AC 2	AC 3
Empennage sized for	AC 1	AC 2	AC 3
Wing sweep (LE) [deg]	36.0	30.1	36.6
Rear spar position [%]	77.0	81.5	81.6
Thickness ratio [%]	8.5	10.1	7.2

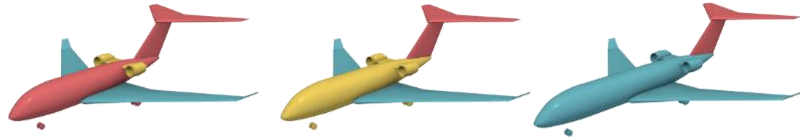
Output metrics

	AC 1	AC 2	AC 3
MTOW [kg]	12000	13060	14940
Block fuel [kg]	3580	4340	5140
DOC [\$ / hr]	12700	14680	16200
Wing area [m ²]	48.9	48.8	59.6
Rated thrust [kN]	20.6	22.4	25.6
Mean DOC [\$ / hr]	14520		
OEM NRC [M\$]	4200		

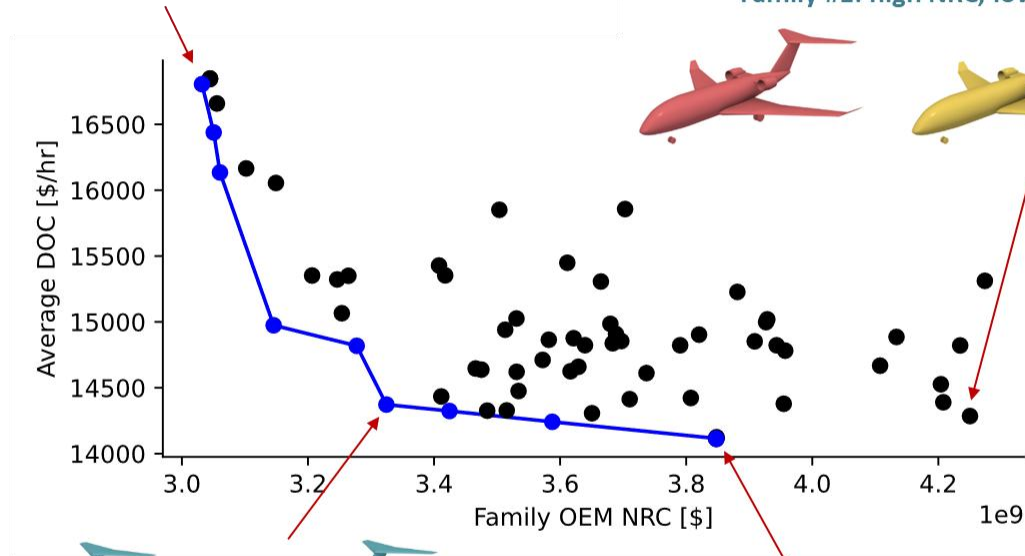
AGILE 4.0
www.agile4.eu

Family concept design

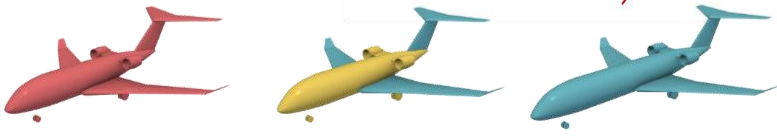
Family #54: lowest NRC, highest DOC (in Pareto front)



Family #1: high NRC, low DOC (no commonality)



Family #30: low NRC, low DOC (good compromise)

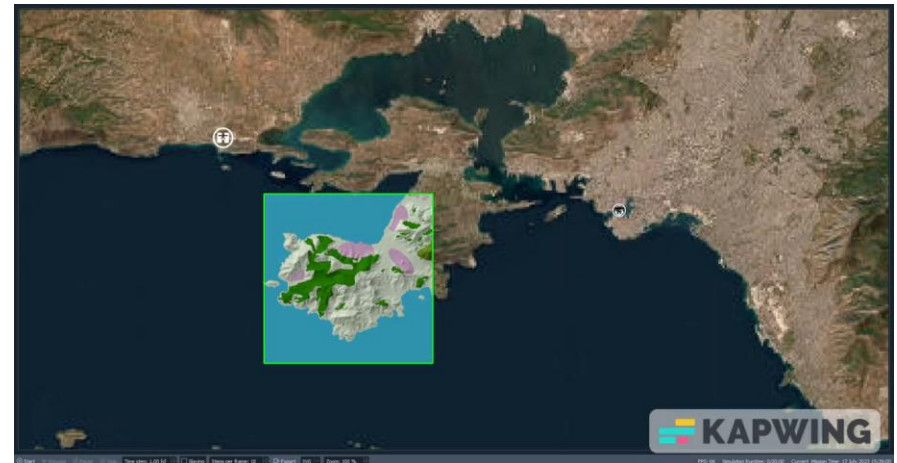
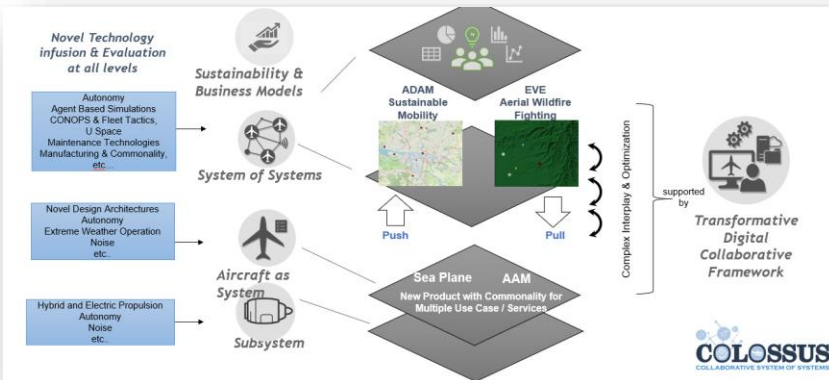


Family #62: highest NRC, lowest DOC (in Pareto front)



System – Of Systems optimization

COLOSSUS project (2023-2026) aims at extended AGILE 4.0 to Systems Of Systems problems with multi levels MDO approach



EVE – Eco Friendly Air Vehicles for Multiple Operating Environment

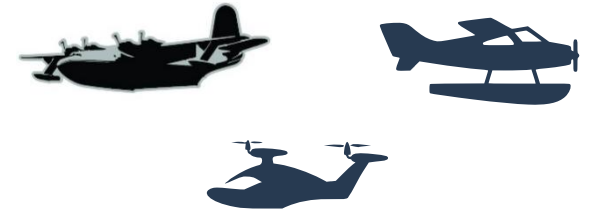
<https://colossus-sos-project.eu/>

Prakas Shiva Prakasha, P., Naeem, N., Amadori, K., Donelli, G., Akbari, J., Nicolosi, F., ... & Nagel, B. (2024). COLOSSUS EU Project–Collaborative SoS Exploration of Aviation Products, Services and Business Models: Overview and Approach. In 34th Congress of the International Council of the Aeronautical Sciences, ICAS 2024. ICAS

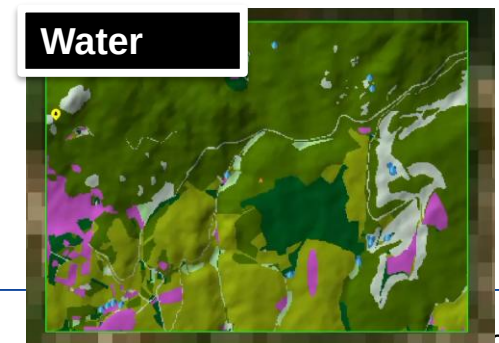
System – Of Systems optimization

Combined optimisation of **composition** and **control** for a SoS problem

- Composition aspects
 - Agents fleet size
 - Agents TLARS (Range / Payload / Speed)
- Control aspects
 - 2 levels of **Tactics** block including “swap” variables
 - Depending on conditions
 - Depending on threshold



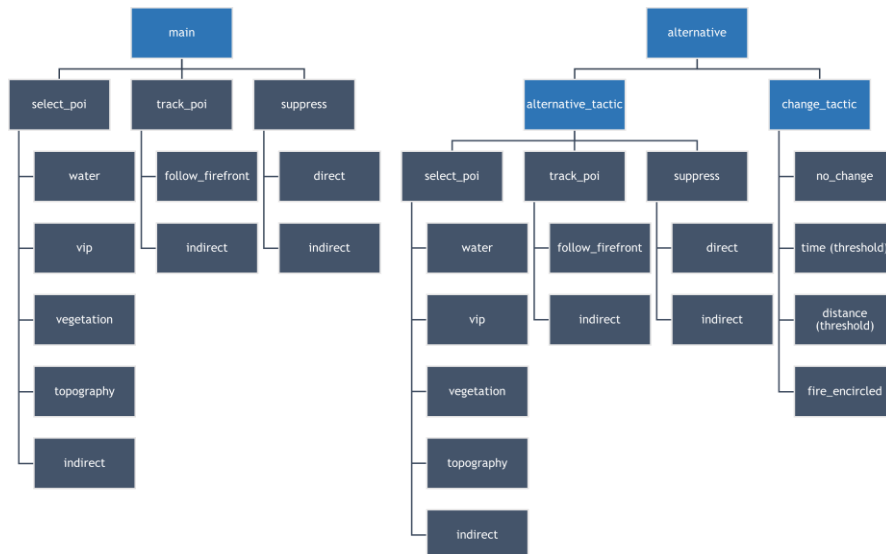
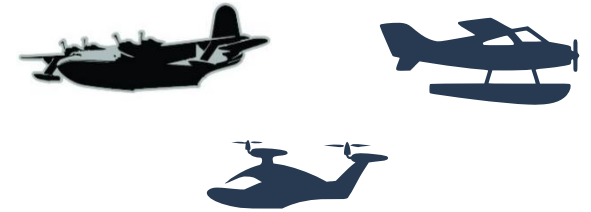
Tactic Name	Acronym	Suppression Target
Water	WAT	Highest spread rate, prioritizes cities
Topography	TOPO	Highest spread rate, prioritizes jumps in elevation
Vegetation	VEF	Highest spread rate, prioritizes highly combustible areas
Indirect	IND	Fire line, starts at highest spread area



System – Of Systems optimization

Combined optimisation of **composition** and **control** for a SoS problem

- Composition aspects
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- Control aspects
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 - Depending on threshold

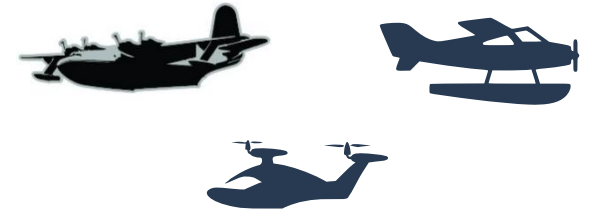


- Ability to change tactics
- Swap from main to alternative
 - Depending on conditions
 - Depending on threshold

System – Of Systems optimization

Combined optimisation of **composition** and **control** for a SoS problem

- Composition aspects
 - Agents fleet size
 - Agents TLARS (Range / Payload / Speed)
- Control aspects
 - 2 levels of **Tactics** block including “swap” variables
 - Depending on conditions
 - Depending on threshold



Catalogue approach

- Selection of **predefined combinations** of main – swap – alternative to choose from

ex: Combination A : indirect attack swap to water if fire reaches 500 meters from residential area

Hierarchical approach

- Tactics degrees of freedom (main, alternative) and swap conditions as **variables**

System – Of Systems optimization

Variables

Seaplane



- Fleet size
- TLAR

eVTOL



- Fleet size
- TLAR

Tactics



Design space

- Main
- Alternative
- Only

System – Of Systems optimization

- **Objectives**

- Fire Cost
- Operation Cost
- Operation Emissions

- **Constraint**

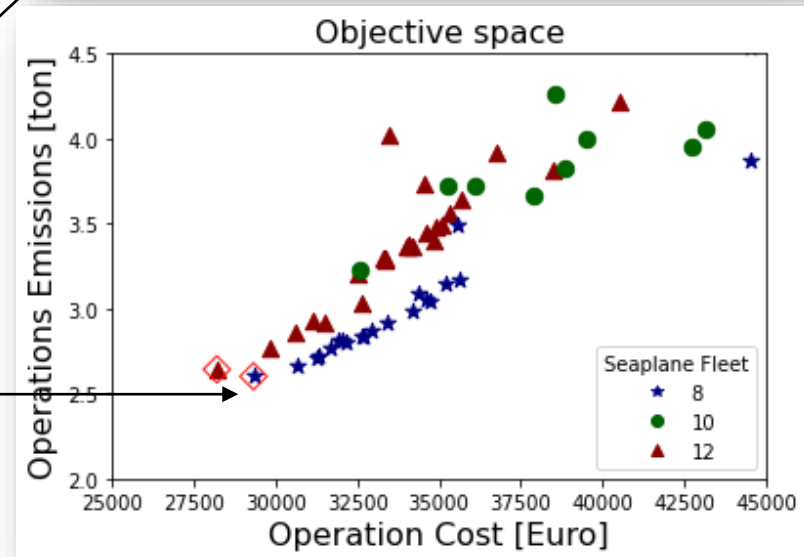
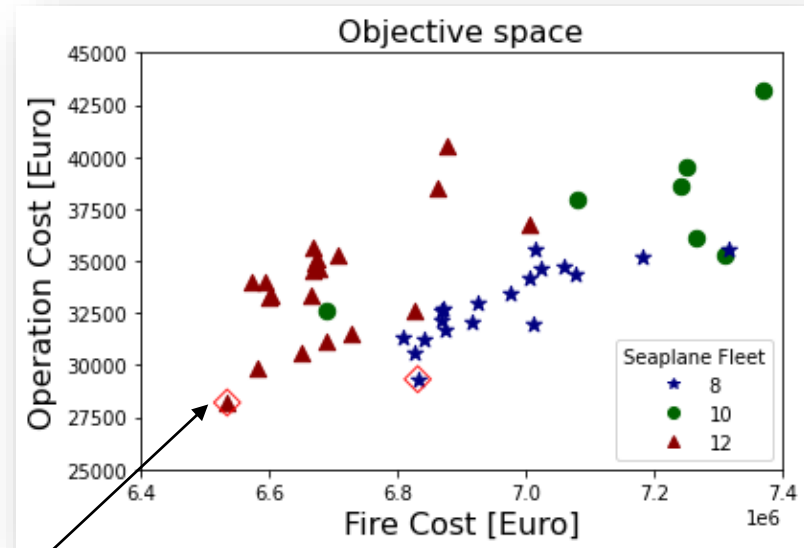
- Mission success

- **Optimization process**

- DOE : 40 Iterations : 180

- **Two main clusters**

- 12-seaplane fleet
→ higher firefighting efficiency
- 8-seaplane fleet
→ lowest operation emissions



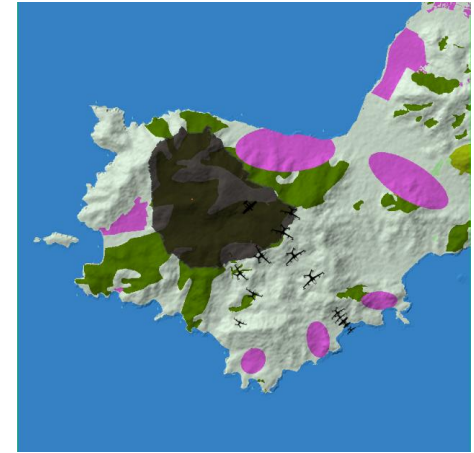
System – Of Systems optimization



Baseline Fleet (2 DHC515)



Best 12-seaplane fleet

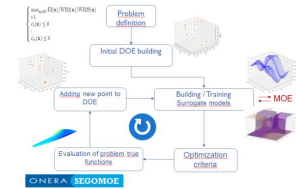


Best 8-seaplane fleet

Table 2: Comparison of the baseline and optimized fleet configurations.

Configuration	Total Fire Cost [M€]	CO ₂ Emissions [t]	Burnt Area [km ²]	Effective Mission Time [h]
Baseline	11.53	1.63	11.79	7.68
8 Seaplanes	6.83	2.60	6.67	2.
12 Seaplanes	6.53	2.64	6.37	1.6

Bayesian optimization: Work in progress



- Multifidelity & Multiobjective Bayesian Optimization

PhD R. Charayron 2020-2023 (ISAE-ONERA)

- Bayesian Optimization with hierarchical variables

PhD P. Saves 2020-2023 (ISAE-ONERA)
Collab J. Bussemaker (DLR)

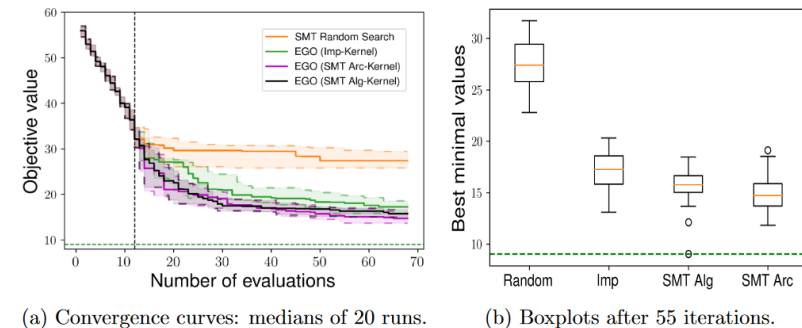
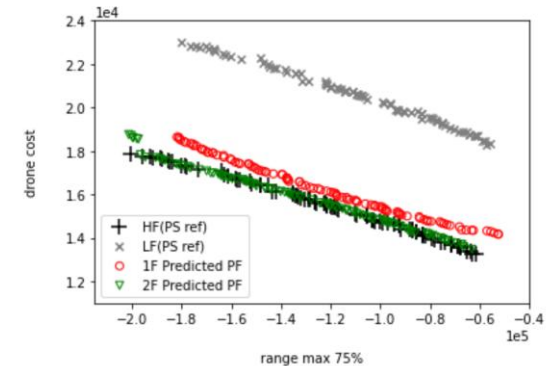
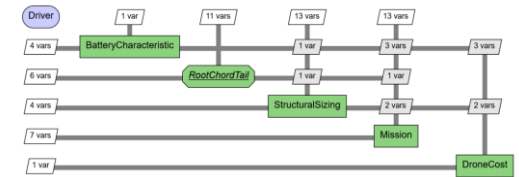


Figure 8: Optimization results for the hierarchical Goldstein function.

Two frameworks

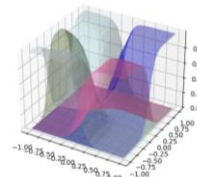


github.com/SMTorg/smt



SMT: Surrogate Modeling Toolbox

- Open source python toolbox: surrogate modeling methods, sampling techniques, and benchmarking functions
- Focus on derivatives (training derivatives used for gradient-enhanced modeling, prediction derivatives)
- New Kriging based surrogate models for higher dimension (KPLS and KPLS-K)
- Noisy Kriging to handle uncertainties on data
- Multifidelity Kriging with or without n MFKPLS)
- Mixture of experts technique for heterogeneous functions
- Mixed integer Kriging to handle **discrete and categorical variables**



SEGOMOE



Mono & multi objective Bayesian optimizer

Mono & Multi fidelity sources

Handling non linear objectives & constraints (black box, no derivative available)

Equality & inequality constraints
(1 ~ 100 constraints)

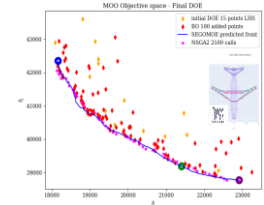
Intermediate dimension problem
(1 ~ 100 variables)

Heterogenous variables (continuous, discrete, categorical)

Costly evaluation (CFD, FEM, objective and/or constraints)

Based on SMT toolbox for surrogate models

Remote access via a **web interface or EGObox**



EGObox - Efficient Global Optimization toolbox



tests passing pytest passing lint passing JOSS 10.21105/joss.04737

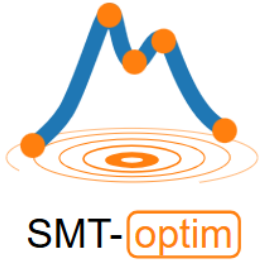
Rust toolbox for Efficient Global Optimization method (arguably the most well-known bayesian optimization algorithm) which addresses the gradient-free optimization of expensive objective functions.

The `egobox` package is twofold:

1. for end-users: [a Python module](#), the Python binding of the optimizer named `Egor` and the surrogate model `Gpx`, mixture of Gaussian processes, written in Rust.
2. for developers: [a set of Rust libraries](#) useful to implement bayesian optimization (EGO-like) algorithms,

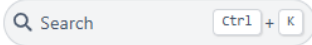
<https://github.com/relf/egobox>

A new open source framework based on SMT



SMT-optim: A Python toolbox for constrained and multi-fidelity Bayesian optimization

SMT-optim is an open-source Python package for Bayesian optimization developed for research applications. It is well suited to expensive-to-evaluate black-box problems that offer limited exploitable structure, such as derivative information. The package supports constrained and multi-fidelity global optimization for mixed-variable design spaces.



SMT-optim Public

Edit Pins Watch 0 Fork 2 Starred 6

main 9 Branches 0 Tags

Go to file Add file Code

oihanc chore: update version number to 0.1.2 ✓	ae35d2a · last week	82 Commits
.github/workflows	test: add auto testing with GitHub Actions	3 weeks ago
docs	fix: improve mfsego numerical stability, add many fidelity lev...	last week
examples	docs: change docs template and structure	2 months ago
src	Merge branch 'feat/improve-benchmarks' into dev	last week
.gitignore	chore: add .gitignore and clean up tracked artifacts	8 months ago
.nojekyll	docs: change docs template and structure	2 months ago
CONTRIBUTING.md	chore: add contributing guidelines	3 weeks ago

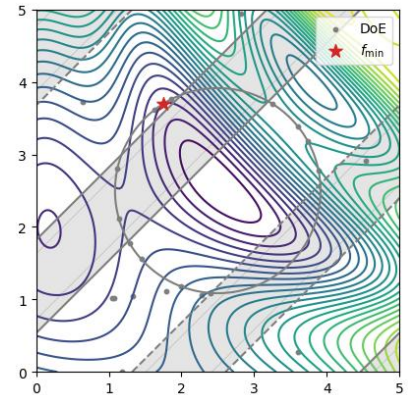
About

SMT optimization

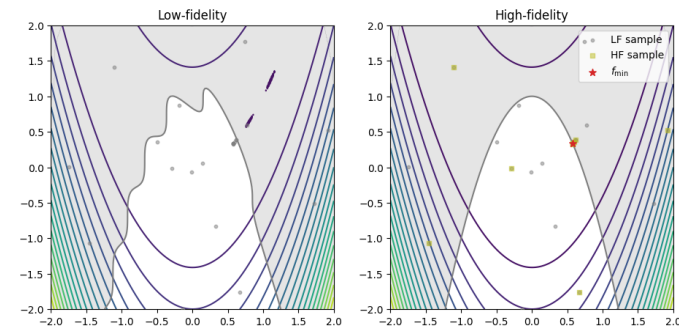
smtorg.github.io/smt-optim/

- Readme
- Apache-2.0 license
- Contributing
- Activity
- Custom properties
- 6 stars
- 0 watching
- 2 forks

Report repository



Sasena test problem with 2 constraints



Multi-fidelity Rosenbrock with 1 constraint

GitHub repository:
Documentation:

<https://github.com/SMTorg/smt-optim>
<https://smtorg.github.io/smt-optim/>

Objectifs du BE

- Partie 1 : Optimisation des hyper-paramètres du GP
- Partie 2 : Mise en place de l'algorithme EGO et comparaison de critères sur un exemple 1D
- Partie 3 : Mise en place de SEGO
- Bonus: optimisation sur un exemple 2D